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Data-driven modeling of amyloid-β targeted antibodies for Alzheimer's disease

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Alzheimer's disease (AD) is characterized by the accumulation of amyloid beta, which is strongly associated with disease progression and cognitive decline. Despite the approval of monoclonal antibodies targeting Aβ, optimizing treatment strategies while minimizing side effects remains a challenge. This study develops a mathematical framework to model Aβ aggregation dynamics, capturing the transition from monomers to higher-order aggregates, including protofibrils, toxic oligomers, and fibrils, using mass-action kinetics and coarse-grained modeling. Parameter estimation, sensitivity analysis, and data-driven calibration ensure model robustness. An optimal control framework is introduced to identify the optimal dose of the drug as a control function that reduces toxic oligomers and fibrils while minimizing adverse effects, such as amyloid-related imaging abnormalities (ARIA). The results indicate that Donanemab achieves the most significant reduction in fibrils. These findings provide a quantitative basis for optimizing AD treatments, providing valuable insight into the balance between therapeutic efficacy and safety.

Alzheimer's disease (AD) is a devastating neurodegenerative disorder and the leading cause of dementia worldwide. Affecting over 47 million people globally, its prevalence is projected to rise significantly in the coming years, placing immense strain on healthcare systems[1]. The disease is characterized by a progressive decline in cognitive function, including memory loss and diminished independence. These symptoms are closely linked to two hallmark pathological features: extracellular Aβ plaques and intracellular neurofibrillary tangles composed of hyperphosphorylated tau protein. Together, these aggregates impair synaptic function, trigger chronic neuroinflammation, and lead to widespread neuronal loss[2–5].

Aβ aggregation plays a central role in the progression of AD. This process begins with the misprocessing of amyloid precursor protein (APP) by β- and γ-secretases, which results in the production of Aβ peptides. These peptides then aggregate into various forms, including toxic oligomers, fibrils, and insoluble plaques[6–8]. Of these, toxic oligomers are considered the most neurotoxic, as they disrupt synaptic communication and are a major contributor to cognitive decline[9–11].

Given the complexity of Aβ aggregation, computational models have become essential tools for understanding the underlying mechanisms and exploring potential therapeutic interventions. Early models focused on the molecular dynamics of Aβ and its spread between neurons, using Smoluchowski equations and kinetic transport models. Later developments introduced spatial dynamics to account for the heterogeneous distribution of misfolded proteins throughout the brain. More advanced models based on partial differential equations (PDEs) now capture the interactions among neurons, astrocytes, and microglia, providing valuable insights into possible therapeutic strategies. Recent innovations have further enhanced these models by incorporating clinical biomarkers and patient-specific parameters[12–15]. Experimental validation plays a crucial role in advancing these computational models. Structural studies offer detailed insights into the molecular aggregation pathways, while kinetic experiments quantify the transition rates between oligomeric and fibrillar states under physiological conditions. These findings continually refine computational models, improving their predictive accuracy and ensuring their alignment with biological observations[2,16,17]. As computational models continue to advance, therapeutic strategies targeting Aβ are also evolving. Monoclonal antibodies, including Aducanumab, Donanemab, and Lecanemab, are designed to reduce plaque burden and slow cognitive decline. While some clinical trials have demonstrated promising results, challenges remain, such as inconsistent efficacy and adverse effects, such as amyloid-related imaging abnormalities (ARIA). In this context, computational models incorporating clinical and spatial data have become crucial tools for optimizing these

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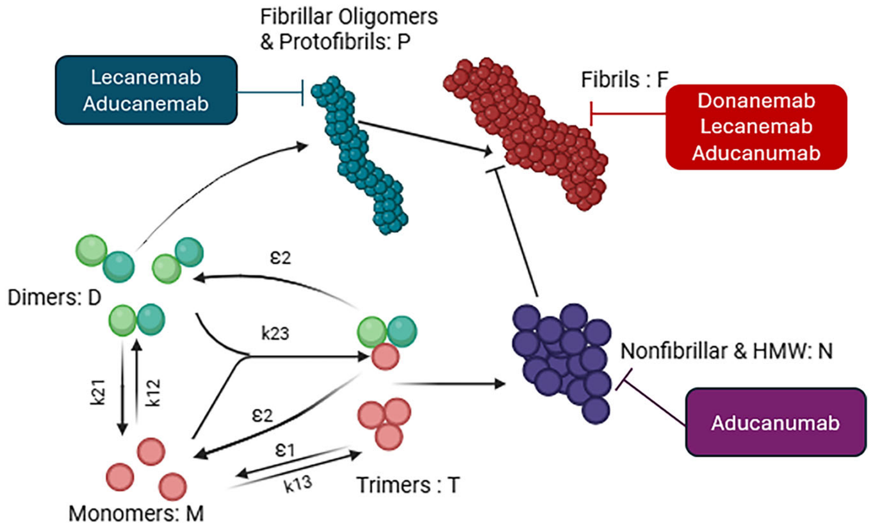
Fig. 1 | The figure illustrates the aggregation of Aβ monomers into oligomers and fibrils, with inhibitory arrows (blunt-ended lines) representing the action of factors that impede the conversion of oligomers to fibrils. The labeled boxes next to each aggregation species indicate the corresponding antibodies involved in their inhibition or clearance. Donanemab, Lecanemab, and Aducanumab target fibrils (F), while Lecanemab and Aducanumab also inhibit fibrillar oligomers and protofibrils (P). Additionally, Aducanumab targets nonfibrillar aggregates and high-molecular-weight species (N). (*Created with BioRender.com*).



therapies[18]. Building upon these advancements, optimal control techniques have increasingly been applied to design treatment strategies for AD, focusing on balancing therapeutic efficacy with minimizing side effects. Recent studies have explored the use of mathematical models incorporating optimal control to personalize anti-Aβ therapy regimens[19,20]. These models leverage patient-specific data to optimize dosing strategies, providing deeper insights into potential therapeutic interventions and helping to tailor treatments for individual patients. However, most existing models focus on late-stage amyloid aggregation, plaque formation, while overlooking the detailed intermediate aggregation process. This gap limits their ability to evaluate early therapeutic interventions targeting toxic oligomers, which are now recognized as key drivers of neurotoxicity. Additionally, current models often prioritize amyloid clearance while neglecting potential treatment side effects, such as ARIA, which remain a major challenge in clinical applications.

To address these gaps, this study develops a comprehensive mathematical framework that captures the full progression of Aβ aggregation from monomers to toxic oligomers, protofibrils, and mature fibrils. By explicitly modeling intermediate species, our approach provides a mechanistic understanding of early aggregation dynamics and their role in AD pathology.

Building on earlier efforts to characterize anti-Aβ antibodies, such as Aducanumab, through in vitro kinetic modeling[21], we expand the scope by incorporating Donanemab and Lecanemab and by integrating in vivo clinical trial data, including plaque reduction and ARIA-E incidence, into our modeling framework. We further introduce an optimal control strategy that balances treatment efficacy and safety, enabling data-driven therapeutic design grounded in both mechanistic modeling and clinical observations.

RESULTS

We developed a mathematical model of Aβ aggregation (Fig. 1) using a system of ordinary differential equations to track its progression from monomers to fibrils, denoted by F; the output of our model represents the final aggregated species and corresponds to Aβ plaque observed in patients.

Sensitivity and identifiability analyses assess key parameters, while uncertainty quantification evaluates model reliability. Finally, we optimize treatment strategies to minimize fibril accumulation and toxic oligomers while reducing side effects, offering potential insights for AD therapy.

Parameter estimation

The values and ranges for 19 key parameters in the Aβ aggregation model, including reaction rates, clearance rates, and saturation values, were estimated through literature review and steady-state analysis. Table 1 summarizes the results, distinguishing previously published values from those newly estimated for this model.

Table 1 | Estimated parameter values and ranges for the mathematical model shown in Eqs. (2)–(8)

Parameters	Description	Value	Unit	Reference
k_{12}	$2M \to D$	800–1180	$M^{-1}s^{-1}$	25
k_{21}	$D \to 2M$	8×10^{-10} – 1.18×10^{-9}	s^{-1}	25
k_{13}	$3M \to T$	$10^{9}(1 \pm \delta)$	$M^{-2}s^{-1}$	Estimated
ϵ_1	$T \to 3M$	10^{-13}–10^{-12}	s^{-1}	8
k_{23}	$M + D \to T$	33–43	$M^{-1}s^{-1}$	25
ϵ_2	$T \to M + D$	10^{-11}–10^{-9}	s^{-1}	8
μ_1	monomers clearance	10^{-5}–10^{-3}	s^{-1}	26
μ_2	dimer clearance	$10^{-8}(1 \pm \delta)$	s^{-1}	Estimated
μ_3	trimer clearance	$10^{-8}(1 \pm \delta)$	s^{-1}	Estimated
μ_4	P clearance	10^{-4}–10^{-3}	s^{-1}	26
μ_5	N clearance	$10^{-4}(1 \pm \delta)$	s^{-1}	26
μ_6	Fibrils clearance	10^{-5}–10^{-4}	s^{-1}	26
$\hat{\mu}_1$	Monomers birth	10^{-14}–10^{-12}	Ms^{-1}	28
λ_1	$\tilde{D} \to P$	10^{5}–10^{7}	$M^{-1}s^{-1}$	Estimated
λ_2	$\tilde{T} \to N$	10^{3}–10^{4}	$M^{-1}s^{-1}$	Estimated
λ_3	$P + N \to F$	0–1	$M^{-1}s^{-1}$	Estimated
K_1	dimer saturation	10^{-7}–4×10^{-6}	M	Assumed
K_2	trimer saturation	10^{-6}–4×10^{-6}	M	Assumed
K_3	N saturation	10^{-3}–5×10^{-3}	M	Assumed

We take $\delta = 0.1$ to define the variability for the corresponding parameters in our study

Sensitivity Analysis

In this study, Sobol sensitivity analysis was used to assess the impact of model parameters on fibril dynamics $F(T)$, our model output at the final time $T = 21.147$ h, in line with experimental data[22]. First-order, second-order, and total sensitivity indices (S_T) were computed across different sample sizes N, using parameter ranges from Table 1. The initial conditions were set as $M(0) = 5\ \mu M$ and all other states at zero[22]. The bar graph in Fig. 2 highlights the most sensitive parameters for $N = 2^{15}$.

To validate the sensitivity indices, Fig. 3 displays the six most sensitive parameters across sample sizes ranging from $N = 2^4$ to $N = 2^{20}$. The x-axis represents $\log_2(N)$, while the y-axis shows the Sobol total sensitivity indices of $F(T)$. The graph demonstrates convergence of these indices.

Additionally, Supplementary Figure 1 presents the total sensitivity indices of $F(t)$ for these parameters at different time points, using a sample size of $N = 2^{15}$.

Table 2 presents Sobol indices and confidence intervals for a sample size of $N = 2^{20}$. The confidence intervals are notably small, typically less than 10% of the corresponding Sobol indices, indicating high accuracy. Their narrow range further highlights the reliability of the estimates.

Figure 4 presents the first, second-order, and total sensitivity analyses, demonstrating consistency across different sensitivity measures. Node sizes represent the first-order Sobol indices (S_i), showing each parameter's direct impact on $F(T)$, while node borders indicate total sensitivity ($S_{T,i}$), capturing both direct and interaction effects. Edge thickness between nodes corresponds to second-order indices ($S_{(i,j)}$), highlighting parameter interactions.

We conducted a Sobol sensitivity analysis on our model under varying initial monomer concentrations, where $M(0)$ represents the initial concentration of the monomer, expressed in micromolar (μM) units. The tested values are:

$$M(0) = 5, 4, 3.5, 3, 2.5, 2, 1.75, 1.5, 1.35, \text{ and } 1.1,$$

over their corresponding parameter ranges with $N = 2^{15}$.

We obtained 10 total Sobol sensitivity indices for each parameter and plotted them in Fig. 5. We set a sensitivity threshold at 10^{-5} to differentiate between sensitive and non-sensitive parameters. Specifically, parameters with average indices below this threshold are considered non-sensitive, while those with indices above it are deemed sensitive. More information and details about the total sensitivity of the model solution under different $M(0)$ values are provided in Supplementary Note 2, as illustrated in Supplementary Figure 2.

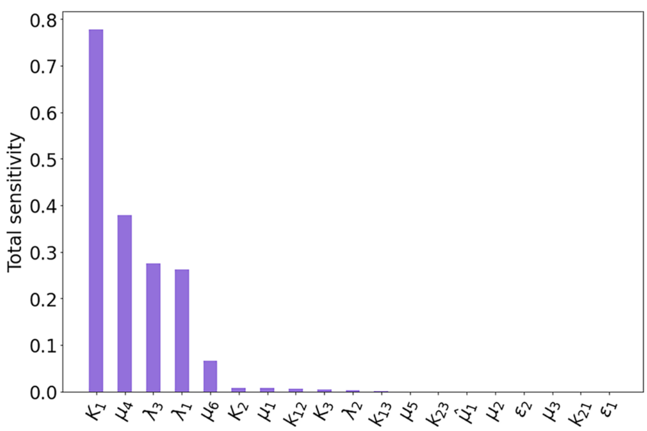


Fig. 2 | Total sensitivity indices of $F(T)$ for different parameters for sample points $N = 2^{15}$. Here, we choose the initial condition $M(0) = 5(\mu M)$.

Data-driven modeling

To calibrate our model of Aβ aggregation (from monomers to fibrils), we used experimental data from[22], which tracked fibril formation over time at ten different starting concentrations of monomers ranging from 5 μM to 1.1 μM. The experiments were conducted under controlled in vitro conditions, where recombinant Aβ42 peptides were incubated in buffered solution at physiological pH and constant temperature. A fluorescent dye, Thioflavin T (ThT), was added to the solution. ThT fluoresces upon binding to β-sheet-rich amyloid fibrils, and its fluorescence intensity increases proportionally with fibril mass. By recording fluorescence at regular time intervals using a plate reader, the researchers monitored the kinetics of fibril formation. After normalization, these curves were scaled to range between 0 and 1 (Fig. 11) and were directly interpreted as fibril concentration over time for use in model fitting and validation. In this section, we calibrate our model using this data and perform parameter identifiability and uncertainty quantification. First, using the parameter ranges specified in Table 1 and a detailed parameter estimation method, we identified a specific feasible region for certain parameters, dependent on the initial monomer concentration ($M(0) = 5, \ldots, 1.1\mu M$). These results are summarized in Table 3.

Next, using normalized experimental data and the Nelder-Mead optimization method, we determined an optimal parameter vector, θ^*, for each of the 10 experimental observations. Practical identifiability classified parameters as identifiable or non-identifiable, with the latter requiring

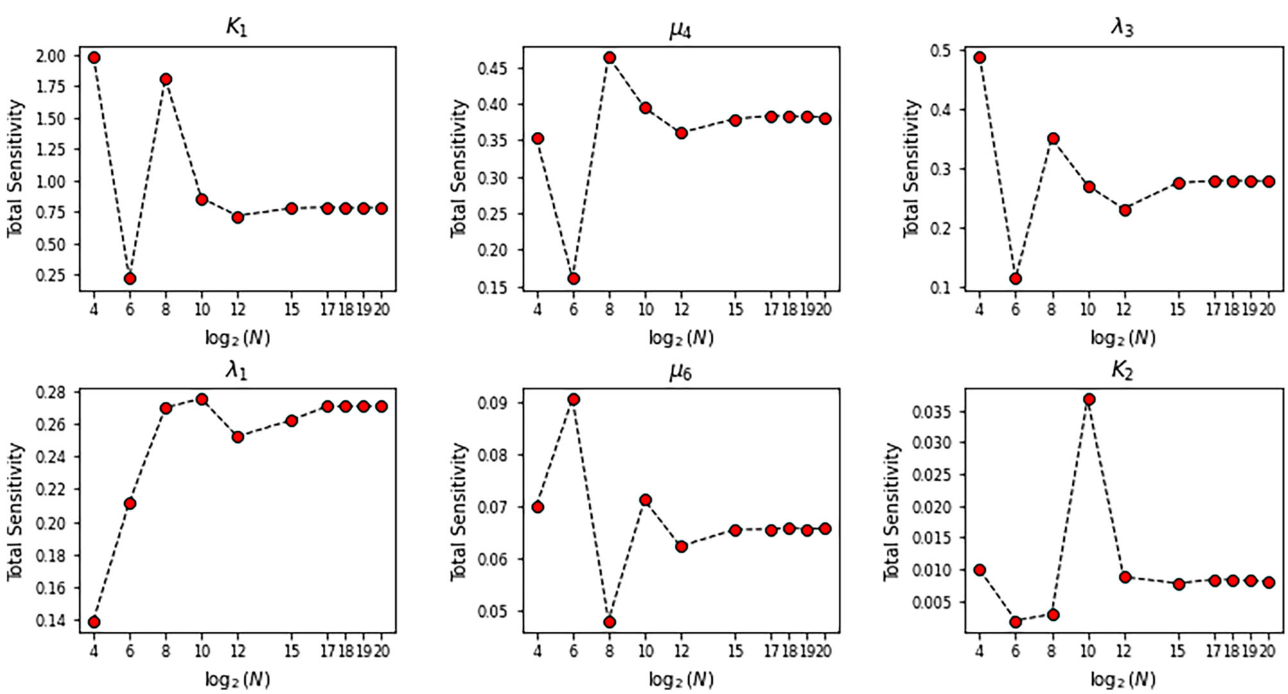


Fig. 3 | Total sensitivity indices of $F(T)$ for the six most sensitive parameters with different sample sizes N and the initial condition $M(0) = 5(\mu M)$.

Table 2 | Total Sobol sensitivity indices (S_T) of $F(T)$ and their corresponding confidence intervals for a sample size of $N = 2^{20}$ across various parameters the initial condition $M(0) = 5(\mu M)$

Parameter	λ_1	λ_2	λ_3	μ_4	μ_5	μ_6	K_1	K_2	K_3	k_{12}
S_T index	2.71e-01	3.06e-03	2.78e-01	3.81e-01	7.07e-05	6.57e-02	7.81e-01	8.00e-03	3.26e-03	5.54e-03
Conf Int	4.93e-03	3.58e-04	4.31e-03	5.62e-03	7.12e-06	1.12e-03	1.33e-02	4.44e-04	2.22e-04	1.38e-04

Parameter	k_{21}	μ_2	μ_1	$\hat{\mu}_1$	ϵ_1	k_{23}	ϵ_2	k_{13}	μ_3
S_T index	4.51e-11	1.07e-09	7.85e-03	1.37e-06	1.34e-16	2.06e-05	8.79e-04	2.33e-10	2.00e-05
Conf Int	1.05e-12	2.36e-11	2.36e-04	5.90e-08	5.95e-17	2.00e-06	2.00e-06	4.16e-11	3.01e-06

The confidence intervals quantify the uncertainty in the estimation of the Sobol indices due to the sampling process in the sensitivity analysis.

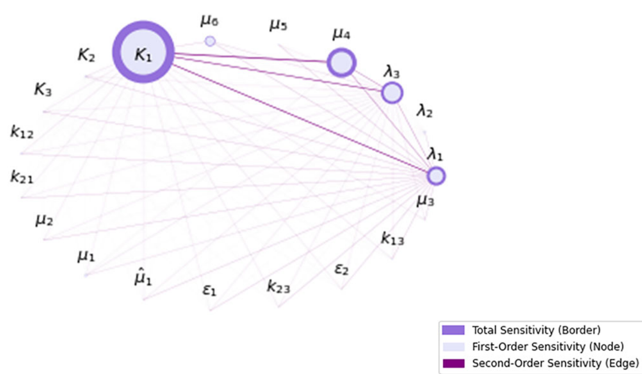


Fig. 4 | Network representation of parameter sensitivities for $N = 2^{15}$, showing first, second, and total order effects on $F(T)$ with $M(0) = 5\mu M$. Node sizes represent the first-order Sobol sensitivity indices (S_i) for each parameter, while node borders highlight each parameter's total sensitivity ($S_{T,i}$), indicating their direct effect on $F(T)$. The thickness of the edges between nodes reflects the second-order sensitivity indices ($S_{(i,j)}$), indicating interaction effects between parameter pairs.

regularization for better model calibration. To handle non-identifiable parameters, we used a regularization strategy that prevented unstable values while keeping a good fit to the data. This approach gently guided poorly constrained parameters toward reasonable values by limiting large deviations from a reference set. As a result, the final optimized parameter vector, $\tilde{\theta}$, remains stable and reliable for each observation. After determining $\tilde{\theta}$ for 10 different initial concentrations of $M(0)$, we summarize the results in Fig. 6.

The detailed values of $\tilde{\theta}$ can be found in Supplementary Table 1 in Supplementary Note 2. Finally, we assess model uncertainty arising from non-identifiable parameters and initial conditions by computing 95% confidence intervals for the model solutions.

Figure 7 displays the concentrations of all state variables for $M(0) = 2\mu M$, using the optimal parameter vector $\tilde{\theta}$. The shaded regions represent 95% confidence intervals, capturing variability from parameter and initial condition uncertainties.

Figure 8 shows the 95% confidence intervals for $F(t)$ across nine additional initial monomer concentrations. The red curves represent model solutions with optimized parameters, while black dots correspond to experimental data. The green-shaded regions indicate the confidence intervals, capturing the uncertainty from non-identifiable parameters and initial conditions.

Molecular targets and mechanisms of anti-$A\beta$ treatments. In our model, each treatment is represented by a control function $u(t)$, which reflects how strongly the drug is acting to clear amyloid plaques over time. Biologically, these functions capture how antibody treatments enhance the removal of F, the final aggregated species in the cascade. This value does not directly reflect the actual drug dose or concentration. The optimization framework then adjusts $u(t)$ over time to balance therapeutic efficacy and side-effect risk. Figure 1 schematically illustrates the molecular targets and mechanisms of Donanemab, Lecanemab, and Aducanumab in the Aβ aggregation cascade. Table 4 summarizes the effectiveness of single and combined anti-Aβ therapies in reducing fibril concentration ($F(t)$). In single-therapy scenarios, each antibody (Donanemab, Aducanumab, or Lecanemab) is administered alone and optimized individually. In contrast, combined therapy refers to the simultaneous administration of antibodies, with their respective control functions co-optimized to enhance overall treatment efficacy. Donanemab (u_D) was the most effective single-drug treatment, achieving over 85% fibril reduction across all initial monomer concentrations ($M(0)$). Lecanemab (u_L) and Aducanumab (u_A) produced moderate reductions of up to 25% and 27%, respectively. Combined therapies, particularly those including all three drugs ("All Drugs"), achieved the highest reductions, exceeding 87% in most cases.

Figure 9 shows the optimal control functions for Donanemab, Aducanumab, and Lecanemab across different initial monomer concentrations $M(0)$, under both single and combined therapy. Although the combination therapy pursues a shared treatment objective, the individual control functions $u_D(t)$, $u_L(t)$, and $u_A(t)$ change differently over time. Each drug plays a distinct role in the optimization process and is subject to its own constraints.

The control functions for single-drug treatments (Donanemab, u_D; Aducanumab, u_A; Lecanemab, u_L) are compared with those from combined therapy scenarios. The solid and dashed lines corresponding to the single-drug and combined therapy are plotted in the same color for each initial condition. From the figure, we can see that when drugs are combined, treatments using Lecanemab (u_L) and Aducanemab (u_A) can be slightly delayed compared to when these drugs are used individually in the single treatment scenario. In our model, the control function represents how much of the maximum effect of the drug is applied over time, not the therapeutic benefit itself. When multiple drugs are used together, the model distributes the treatment effort across all drugs to use them more efficiently. This means that even if one drug starts later or is applied at a lower level, the overall therapeutic effect remains strong because of the combined action. As shown in Table 4, this coordinated timing improves the reduction of amyloid plaque size. The pattern for Donanemab (u_D) remains nearly unchanged between single and combined treatment scenarios. This is expected because Donanemab appears to be the dominant treatment, likely due to its higher maximum efficacy $U_{\max}^D$ compared to the other drugs.

Figure 10 illustrates the robustness of fibril reduction across different initial monomer concentrations. The 95% confidence intervals for fibril concentrations ($F(t)$) following single-drug treatments highlight the model's reliability under varying initial conditions.

Discussion

This study develops a mathematical model of $A\beta$ aggregation, focusing on its transition from monomers to fibrils. The model integrates differential equations, mass-action kinetics, and coarse-grained modeling to provide a detailed view of the aggregation process (Fig. 1). Sensitivity analysis, parameter estimation, and uncertainty quantification confirm the model's robustness and reliability, offering insights into AD progression and potential treatment strategies.

By identifying 19 key parameters that govern $A\beta$ aggregation, our parameter estimation and sensitivity analyses ensure that reaction rates,

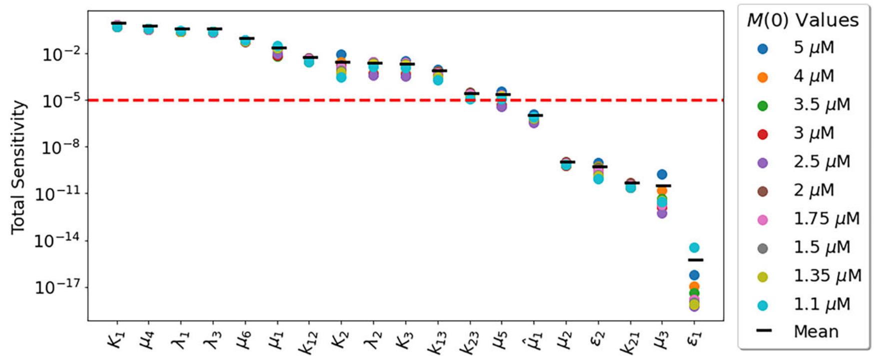


Fig. 5 | Sobol total sensitivity indices for each parameter with 10 different initial conditions, showing total sensitivity indices for each initial condition (dot plot) and the average value (a dashed line). The sensitivity threshold 10^{-5} is also used to divide sensitive and non-sensitive parameters.

Table 3 | Parameter ranges for different initial monomer concentrations

Initial Conditions	Parameter Ranges
$M(0) > 2\mu M$	$K_1 = [10^{-7}, 4 \times 10^{-6}]$
	$K_2 = [10^{-6}, 4 \times 10^{-6}]$
	$\lambda_2 = [10^3, 10^4]$
$M(0) \leq 2\mu M$	$K_1 = [10^{-7}, 2 \times 10^{-6}]$
	$K_2 = [10^{-6}, 2 \times 10^{-6}]$
	$\lambda_2 = [10^4, 10^5]$

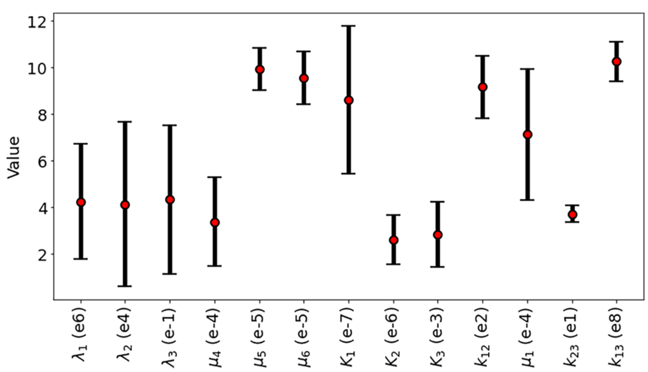


Fig. 6 | Statistical distribution of the optimized parameter values across different initial monomer concentrations ($M(0)$). Red points indicate the mean of the optimized parameters ($\tilde{\theta}$) across the 10 conditions, with error bars representing standard deviations. The numerical values in parentheses specify the means and standard deviations for each parameter.

clearance mechanisms, and saturation effects align with experimental data. The consistency of Sobol sensitivity indices across different sample sizes further supports the model's stability. Additionally, sensitivity analysis across different initial monomer concentrations ($M(0)$) highlights the model's adaptability to biologically relevant conditions, making it useful for optimizing treatment strategies under various physiological settings. Parameters such as k_{13} and λ_1, which govern trimer and protofibril formation, showed high sensitivity indices due to their strong influence on Aβ dynamics. In contrast, parameters like ϵ_1 (the rate of trimer-to-monomer conversion) and μ_5 (the clearance rate of high molecular weight (HMW) oligomers) displayed much lower sensitivity, suggesting minimal or redundant contributions to the model structure.

Calibrating the model with experimental data highlights the importance of data-driven methods in improving both predictive accuracy and interpretability. Using the parameter ranges from Table 1 and advanced estimation techniques, we identified feasible ranges for key parameters across different initial monomer concentrations ($M(0)$), as shown in Table 3. This calibration ensures that the model's predictions align well with experimental observations. Parameter identifiability analysis, using experimental data, showed that certain reaction rates and clearance parameters were identifiable. Uncertainty quantification confirmed the reliability of the parameter estimates, showing that aggregation pathway predictions remain consistent across experimental conditions.

Notably, the model underestimates the lag phase and the steep rise in fibril formation at low monomer concentrations, most likely because it does not incorporate secondary nucleation or feedback mechanisms. Mechanistically, secondary nucleation involves a feedback process in which existing fibrils catalyze the formation of new oligomeric species. Capturing this pathway in our model would require linking the coarse-grained fibrillar compartment back to the earlier mass-action system, effectively coupling two distinct modules in a bidirectional manner. This would introduce additional kinetic parameters and backward fluxes that are difficult to estimate reliably without more detailed experimental data. Given these limitations, we chose not to include secondary nucleation to avoid overfitting and identifiability issues, particularly in light of sparse data and varying initial conditions. With six ODEs and nineteen parameters, the model is already complex, and we prioritized interpretability to support treatment simulation and control optimization. This simplification also helps preserve numerical and structural stability, which is important for reliable performance in downstream therapeutic applications. Nonetheless, secondary nucleation remains an important biological mechanism, and future models may incorporate it.

By integrating experimental data with theoretical predictions, this data-driven approach enhances the precision and applicability of the Abeta aggregation model, providing a strong framework for understanding AD and optimizing treatments. The treatment of AD has advanced with monoclonal antibodies like Donanemab, Lecanemab, and Aducanumab, each targeting different forms of $A\beta$ plaques. Donanemab specifically binds to $A\beta_{pE3}$, a modified form of fibrils found in plaques. While our model does not distinguish fibril subtypes like $A\beta_{pE3}$, it uses F, the final aggregated form, to reflect the aggregated species targeted by these therapies, consistent with their clinical function. Lecanemab targets protofibrils and plaque fibrils, while Aducanumab binds both soluble oligomers and fibrils. Although Aducanumab is no longer in clinical use due to an unfavorable risk-benefit profile, it is included in our model for comparative purposes.

To evaluate treatment safety, we incorporate a side-effect function focused on ARIA-E, the most commonly reported adverse effect, which shows a clear dose-dependent relationship with treatment. Although ARIA-H is not dose-dependent and often occurs in patients with existing ARIA-E, summing both ARIA-E and ARIA-H would overestimate total ARIA incidence, as some patients experience both subtypes, and their overlap is not well defined in clinical trial data. Therefore, we focus on ARIA-E to more accurately capture the dose-dependent nature of side effects. By simulating different treatment strategies, our model aims to optimize the balance between efficacy and safety. To study the effects of each drug, we incorporated them as control functions in our mathematical model. This helped us

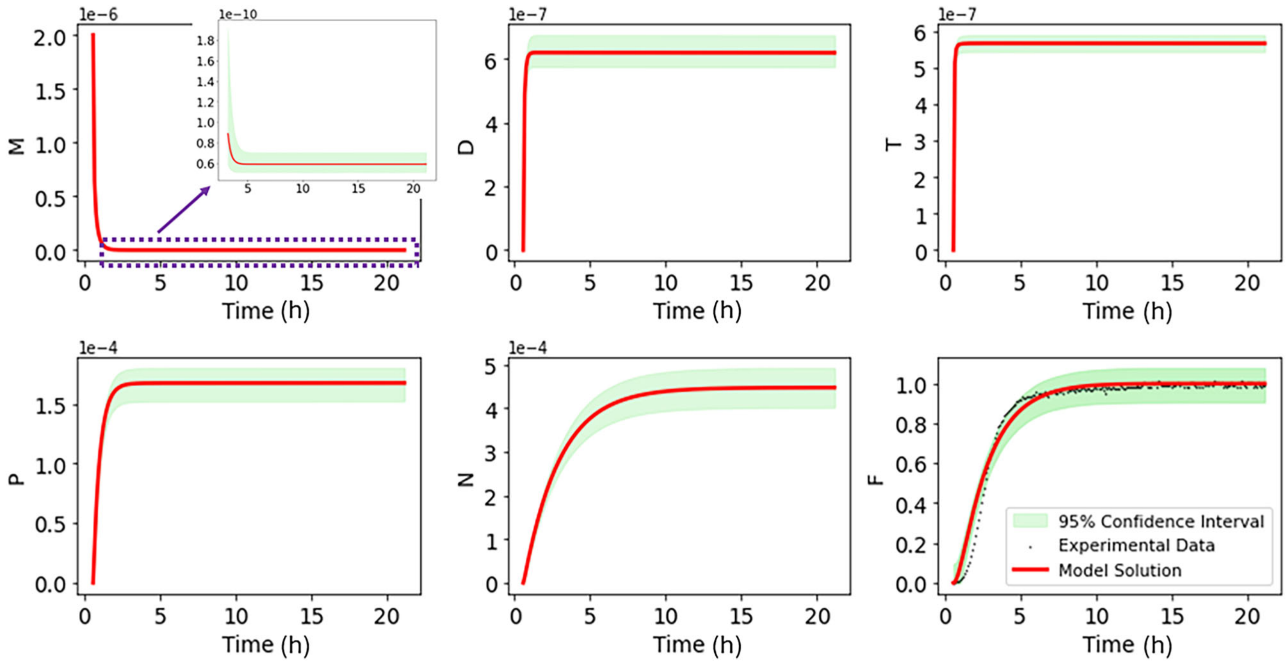


Fig. 7 | Concentration of all state variables with the initial concentration $M(0) = 2\mu M$. The shaded areas represent the 95% confidence intervals.

determine the best timing and dosage for treatment. The model also tracks all key stages of Aβ aggregation, including protofibril, soluble oligomer, and fibril formation, giving a complete picture of how these therapies influence disease progression.

We acknowledge that $A\beta$ aggregate species exist along a continuum rather than as sharply defined entities. Their classification is typically based on operational definitions using empirical separation and detection assays such as size-exclusion chromatography (SEC), electron microscopy (EM), or dot blot. Dot blot assays use antibodies to detect specific conformations or sizes of aggregates directly on a membrane, but they do not provide structural resolution. These methods yield assay-dependent categorizations that can vary across studies. In our modeling framework, we adopt the aggregate labels and definitions as reported in the primary experimental literature, while fully recognizing the inherent limitations and uncertainty in categorizing these species.

Our results show that Donanemab is the most effective single treatment for reducing fibrils, achieving over 85% reduction across all initial conditions. When combined with Lecanemab and Aducanumab, the reduction improves slightly to over 87%, but this difference is not statistically significant. A key advantage of using Lecanemab and Aducanumab together is that their treatment timing can be more flexible, allowing for a slower titration. Our study offers valuable insights into personalizing AD treatment. By simulating different starting conditions and treatment plans, our model shows how therapies can be customized for each patient. We also included a side-effect function based on clinical data, helping us balance the benefits of treatment with potential risks like ARIA, ensuring safety alongside effectiveness. While our model provides a solid foundation for optimizing Aβ-targeting therapies, there's room for improvement. Incorporating patient-specific pharmacokinetics could make predictions more accurate and support better personalized strategies. Including additional side effects, such as immune responses or chronic inflammation, would give a more complete picture of treatment risks. In addition, we did not account for the direct interactions among Aducanumab, Lecanemab, and Donanemab, which remain poorly understood. Combining these therapies could theoretically increase the risk of side effects while potentially enhancing efficacy. Further research is needed to clarify the potential benefits and risks of combination therapy.

Beyond Aβ-targeting therapies, this modeling framework could be applied more broadly. Adding biomarkers for neuroinflammation or tau protein would help us better understand AD progression, while exploring combination therapies, like monoclonal antibodies paired with anti-tau drugs, could improve treatment outcomes. Finally, this framework could be extended to other neurodegenerative diseases characterized by protein aggregation. Its flexible, data-driven foundation offers a versatile platform for advancing precision medicine in diverse protein-folding disorders.

Methods

The aggregation of Aβ from monomers to fibrils is a complex and multi-step process involving several intermediate states. Initially, soluble Aβ monomers, which are primarily in random coil or α-helix conformations, associate to form dimers and trimers through non-covalent interactions[4]. These small oligomers can further aggregate into larger fibrillar oligomers with β-sheet structures, leading to the formation of protofibrils, which are elongated precursors to mature fibrils[23]. Some oligomers, however, remain as non-fibrillar and HMW oligomers, which are less structured but highly toxic[6]. Eventually, the protofibrils elongate and stack to form insoluble, stable fibrils characterized by a cross-β sheet structure. These fibrils accumulate as amyloid plaques in the brain, a hallmark of AD[4]. Understanding these intermediate stages is crucial for developing interventions to prevent or disrupt amyloid aggregation[23]. In Figure 1, we summarize the abeta aggregation process[24] and define the following variables to categorize the various species of Aβ aggregates:

- M: Monomers
- D: Dimers
- T: Trimers
- P: Fibrillar oligomers and Protofibrils
- N: Non-fibrillar oligomers and HMW oligomers
- F: Fibrils

We propose a system of differential equations to develop a mathematical model that describes the dynamics of Aβ aggregation, as illustrated in Figure 1. Then we model this process by following two principal parts below:

- **Mass Action Kinetics:** This traditional approach will be applied to the formation and interaction dynamics of the lower molecular weight species M, D, and T. The mass action law provides a framework to quantify how these species combine or convert from one form to another over time.

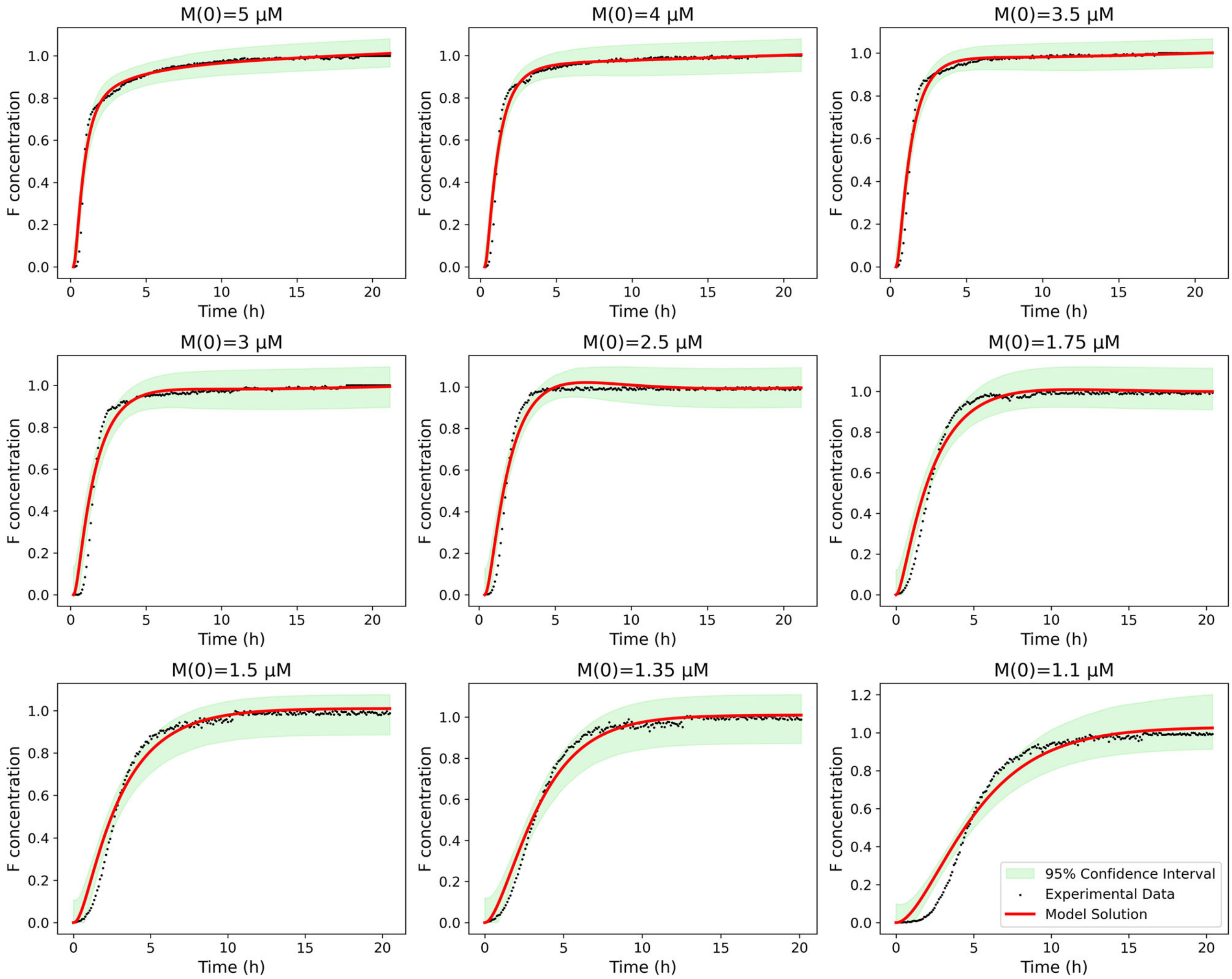


Fig. 8 | Confidence intervals for nine different initial concentrations of $M(0)$, accounting for uncertainties introduced by non-identifiable parameters. Red curves represent the simulated solutions using the optimal parameter vector $\tilde{\theta}$, while black dots correspond to experimental data. The 95% confidence intervals (green-shaded regions) reflect how uncertainties in non-identifiable parameters influence model predictions across different initial conditions.

Table 4 | Percentage reduction in fibril concentration, computed using the formula $\frac{F(T)-F_u(T)}{F(T)} \times 100$ for various initial monomer concentrations

$M(0)$ Treat	5μM	4μM	3.5μM	3μM	2.5μM	2μM	1.75μM	1.5μM	1.35μM	1.1μM
D	85.7	85.8	85.9	86.1	85.9	86.0	86.0	85.9	85.9	85.8
L	15.6	17.6	17.9	20.7	20.8	16.0	15.5	16.1	20.6	24.9
A	17.0	19.1	19.2	22.4	24.5	14.4	12.1	17.4	21.9	27.0
A and L	28.6	31.9	32.1	36.8	41.1	26.5	24.0	29.2	36.2	43.2
A and D	86.5	86.9	86.9	87.4	87.8	86.1	86.3	86.6	87.3	88.2
L and D	86.4	86.6	86.9	87.4	88.1	86.6	86.6	86.7	87.3	88.0
All drugs	87.1	87.7	87.8	88.6	89.4	86.8	86.9	87.3	88.5	89.1

Here, $F(T)$ denotes the fibril concentration at the final time without treatment, while $F_u(T)$ represents the fibril concentration at the final time under the corresponding treatment.

- **Coarse-Grained Modeling:** For the higher molecular weight complexes P, N, and F, a coarse-grained approach is adopted. This method allows us to represent the aggregation behavior of these larger complexes more abstractly, which is useful in capturing the essential features of their dynamics without detailing every underlying interaction.

This newly developed model in this paper aims to comprehensively describe the aggregation pathways and kinetics of Aβ species, facilitating deeper insights into the mechanisms driving the progression of amyloid-related diseases and exploring the efficacy of possible treatments.

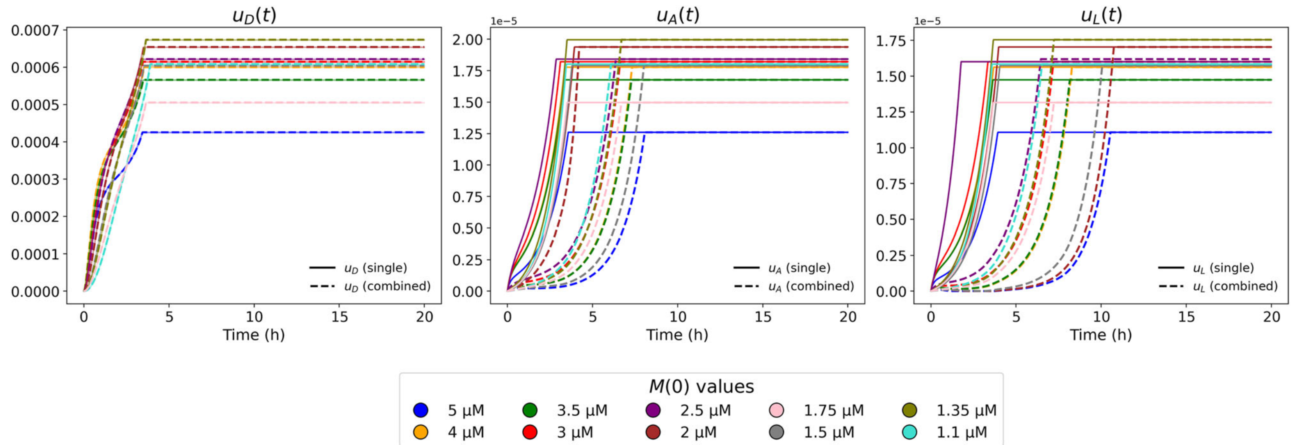


Fig. 9 | Optimal control functions for all initial monomer concentrations $M(0)$, comparing single drug treatments (solid lines) with combination therapy (dashed lines). The control functions (left: Donanemab, center: Aducanumab, right: Lecanemab) represent how strongly each drug acts to clear amyloid plaques over time. Plateaus indicate periods when the drug is applied at full effect.

The mass action model

To accurately model the kinetics of Aβ aggregation, we consider the fundamental chemical reactions among monomers (M), dimers (D), and trimers (T). These reactions are pivotal in understanding the aggregation pathway that leads to amyloid fibril formation. Below, we present the primary reactions and their corresponding rate constants:

$$2M \underset{k_{21}}{\overset{k_{12}}{\rightleftharpoons}} D,\ 3M \underset{\epsilon_1}{\overset{k_{13}}{\rightleftharpoons}} T, \quad M + D \underset{\epsilon_2}{\overset{k_{23}}{\rightleftharpoons}} T. \tag{1}$$

Using the general formula of mass action provided in Supplementary Note 1, we can derive the following equations for monomers, dimers, and trimers that describe these chemical reactions.

The equation of Monomers. The dynamics of the monomer concentration are influenced by various processes, presented in the following differential equation:

$$\begin{aligned} \tfrac{d[M]}{dt} = \hat{\mu}_1 - k_{12}[M]^2 + k_{21}[D] - k_{13}[M]^3 \\ +\epsilon_1[T] - k_{23}[M][D] + \epsilon_2[T] - \mu_1[M]. \end{aligned} \tag{2}$$

There are several key terms in this equation including: $k_{12}[M]^2$ describes the aggregation of monomers to form dimers; $k_{21}[D]$ models the dissociation of dimers into monomers; $k_{13}[M]^3$ represents the aggregation of monomers to form trimers; $\epsilon_1[T]$ means the conversion of trimers back to monomers; $k_{23}[M][D]$ describes the interaction between monomers and dimers to form trimers; and $\epsilon_2[T]$ is the conversion of trimers back to monomers and dimers. Here, $\hat{\mu}_1$ denotes the external production of Aβ monomers through enzymatic cleavage of APP, and $\mu_1[M]$ represents biological clearance mechanisms such as enzymatic degradation (e.g., by neprilysin), microglial uptake, or efflux across the blood-brain barrier. These terms are included to account for the physiological turnover of monomers independent of aggregation dynamics.

The equation of Dimers. The rate of change in dimer concentration is given by:

$$\frac{d[D]}{dt} = k_{12}[M]^2 - k_{21}[D] - k_{23}[M][D] + \epsilon_2[T] - \mu_2[D], \tag{3}$$

where $\mu_2[D]$ represents the degradation rate of dimers.

The equation of trimers. Finally, the rate of change in trimer concentration is described by:

$$\frac{d[T]}{dt} = k_{13}[M]^3 - \epsilon_1[T] + k_{23}[M][D] - \epsilon_2[T] - \mu_3[T], \tag{4}$$

where $\mu_3[T]$ is the clearance of trimer.

The coarse-grained model

To model the dynamics of larger molecular weight species in Aβ aggregation, we employ a coarse-grained approach as outlined in the following reaction scheme:

$$\alpha D \to P,\ \beta T \to N,\ P + N \to F \tag{5}$$

This model simplifies the complex interactions into direct pathways leading to the formation of larger complexes, thereby enabling a focus on the essential aspects of their behavior. We introduce the coefficients α and β, which represent the numbers of dimers (D) and trimers (T) to form fibrillar oligomers and protofibrils (P), as well as non-fibrillar oligomers and HMW oligomers (N), respectively. In this case, we introduce $\tilde{D} = \alpha D$ and $\tilde{T} = \beta T$, the formations of P and N are directly tied to the concentrations of $\tilde{D}$ and $\tilde{T}$, respectively.

Equation for fibrillar oligomers and protofibrils. The concentration of P is described by the following logistic growth model:

$$\frac{d[P]}{dt} = \tilde{\lambda}_1[\tilde{D}](\tilde{K}_1 - \tilde{D}) - \mu_4[P] = \tilde{\lambda}_1[\alpha D](\alpha K_1 - \alpha D) - \mu_4[P],$$

which implies that

$$\frac{d[P]}{dt} = \lambda_1[D](K_1 - [D]) - \mu_4[P], \tag{6}$$

where $\lambda_1 = \alpha^2\tilde{\lambda}_1$ and K_1 represents the carrying capacity of D.

Equation for non-fibrillar oligomers and HMW oligomers. Similarly, the dynamics of N are modeled by:

$$\frac{d[N]}{dt} = \lambda_2[T](K_2 - [T]) - \mu_5[N], \tag{7}$$

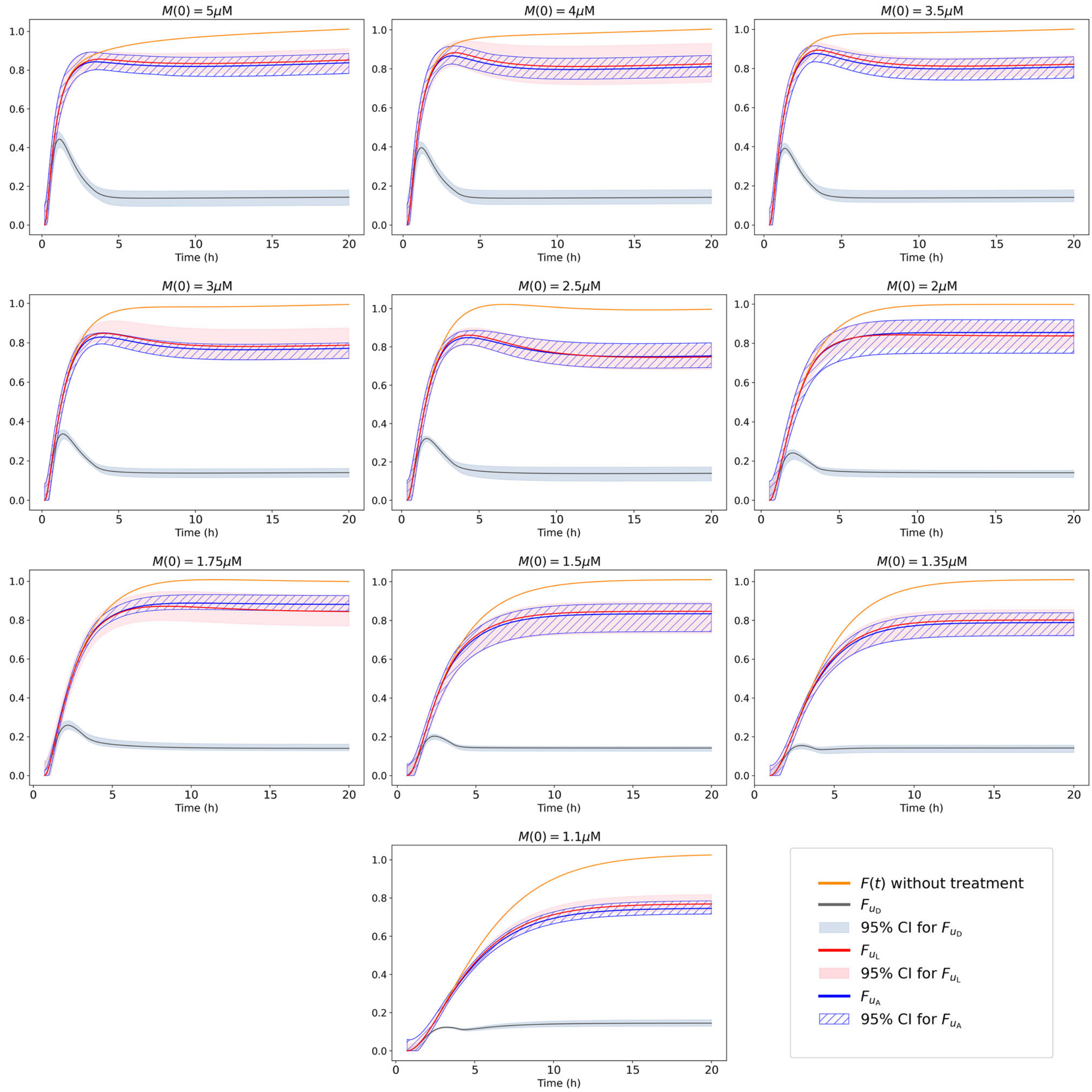


Fig. 10 | 95% confidence intervals for F after single-drug treatments with u_D, u_L, and u_A (shaded in gray, pink, and blue, respectively). The orange curve represents the F concentration without treatment, plotted for ten different initial concentrations of $M(0)$.

where K_2 is the saturation limit of T, reflecting the maximum feasible concentration.

Equation for Fibrils. The formation of fibrils (F) is influenced by both P and N, with the following Michaelis-Menten Kinetics:

$$\frac{d[F]}{dt} = \lambda_3 \frac{[P]}{1 + [N]/K_3} - \mu_6[F]. \tag{8}$$

Parameter Estimation

Estimating parameter values is essential for ensuring that mathematical models accurately reflect observed data. Whenever possible, we use literature-based values, as they are well-documented and reliable. When unavailable, a range of values is considered. For reactions covered in previous studies, we adopt existing parameter ranges, while for new reactions unique to our model, we estimate parameters using steady-state concentrations. Throughout this paper, aggregate concentrations are reported in molar (M) or micromolar (μM) units.

Parameters for the mass action model. In this subsection, we consider the monomer-dimer-trimer process described by Equations (2), (3), and (4). The law of Mass Action is utilized and then taking into account existing literature and the steady-state concentrations, we determine the parameter ranges as follows:

Parameters in Eq. 2. From literature: We take the range of k_{12} as $800 - 1180(M^{-1}s^{-1})$ and k_{23} as $33 - 43$ $(M^{-1}s^{-1})$ from[25] since our model closely resembles the monomer-dimer-trimer process outlined by[25]. We take the range of the death rate, μ_1, as $10^{-5} - 10^{-3}$ (s^{-1})form[26] (consistent with $\mu_1 = 10^{-5}$ (s^{-1}) in[27]) and the birth rate, $\hat{\mu}_1$, as $(10^{-14}, 10^{-12})$ (Ms^{-1}) from[28]

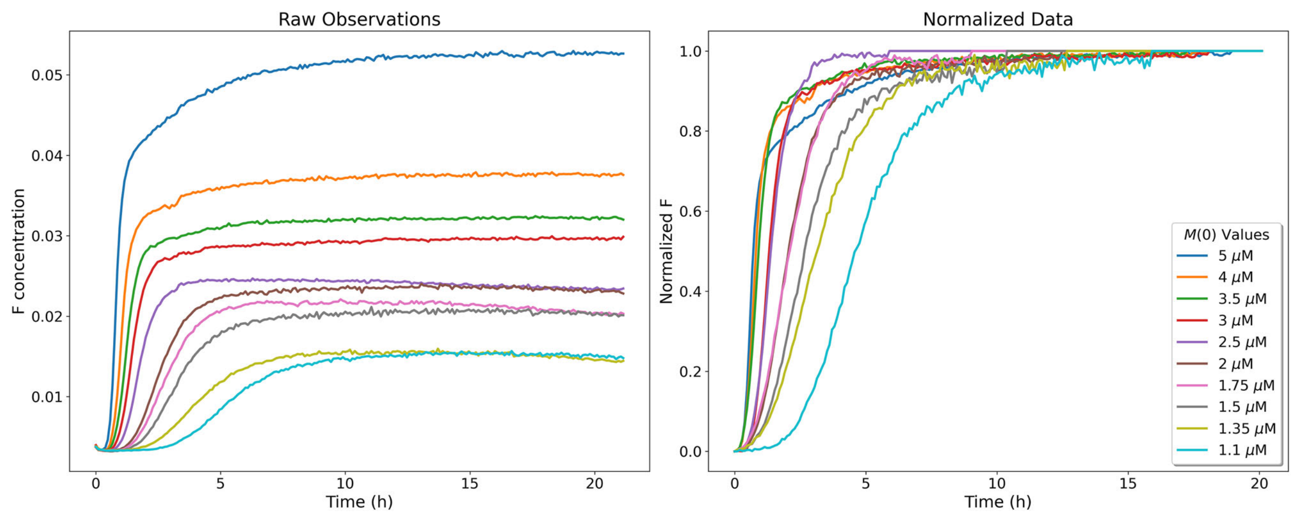


Fig. 11 | Raw experimental data from[22] for the model output F, alongside the normalized F, representing the longitudinal concentrations of $F(t)$ with 10 different initial conditions $M(0)$.

(consistent with $\hat{\mu_1} = 10^{-12}\,(Ms^{-1})$ in[27]). Since the models in the literature lack considering the reverse of certain reactions, we take ϵ_1 within the range $(10^{-13}, 10^{-12})\,(s^{-1})$.

From steady-states: The early-stage interactions among monomers, dimers, and trimers are modeled using mass-action kinetics. In the absence of downstream aggregation, this closed system conserves total Aβ concentration, and the steady-state levels of dimers and trimers remain bounded by the initial monomer supply (1.1-5 μM), typically not exceeding 10^{-6} M. Consequently, the monomer concentration should decrease; therefore, we assume that the concentrations of M and D in the steady-state situation are $[M^\infty] \approx 10^{-9}\,(M)$ and $[D^\infty] \approx 10^{-6}\,(M)$. By considering on the steady-state of the reaction between M and D only, we have $k_{21} = \frac{k_{12}[M^\infty]^2}{[D^\infty]} = (8 \times 10^{-10}, 1.18 \times 10^{-9})\,(s^{-1})$.

Parameters in Eq. 3. In this equation, the value of k_{23} remains the same as previously defined. We have already set k_{12} and k_{21} in the previous equations to maintain the steady state. By using the steady-state condition, we have $\epsilon_2[T] \approx k_{12}[M]^2$, which implies $\epsilon_2 \approx 10^{-9}\,(s^{-1})$. Since the conversion of the trimer back to monomer and dimer is negligible[8], we choose a range of $(10^{-11}, 10^{-9})\,(s^{-1})$ for ϵ_2. We choose the same range for μ_2 as for μ_3 since the clearance rates of the dimer and trimer are similar.

Parameters in Eq. 4. By considering the steady-state of Eq. (4) and neglecting $\epsilon_2[T^\infty]$ which is nearly zero, we have:

$$k_{13} = \frac{\epsilon_1[T^\infty]}{[M^\infty]^3} \; and \; \mu_3 = \frac{k_{23}[M^\infty][D^\infty]}{[T^\infty]}.$$

By choosing $\epsilon_1 \approx 10^{-12}\,(s^{-1})$, we have $k_{13} \approx 10^9\,(M^{-2}s^{-1})$. Similarly, by setting $k_{23} \approx 40\,(M^{-1}s^{-1})$, we determine $\mu_3 \approx 10^{-8}\,(s^{-1})$.

The Coarse-Grained model. In this subsection, we attempted to establish a range for the parameters appearing in Equations (6), (7), and (8). Since these parameters are not directly reported or utilized in other literature, we initially identified an acceptable range for them.

From Literature: The degradation rates μ_4, μ_5, and μ_6 are reported in refs. 26–28 and range from 10^{-7} to $10^{-3}\,(s^{-1})$.

Parameters in Eq. 6. The concentration of dimers can reach around $10^{-6}\,(M)$, we take the carrying capacity of D, K_1, to be in a range of $(10^{-7}, 4 \times 10^{-6})\,(M)$. Since P is formed by the accumulation of many dimers, $[P^\infty]$ is much greater than $[D^\infty]$. On the other hand, P combines with N to form fibrils, the concentration of which at steady state is around $10^{-2}\,(M)$ according to experimental data[22]. Therefore, we assume $[P^\infty] = (10^{-4}, 10^{-3})\,(M)$. Additionally, we assume $\mu_4 = (10^{-4}, 10^{-3})\,(s^{-1})$ and have $[D^\infty](K_1 - [D^\infty])$ vary between 10^{-14} and $10^{-12}\,(M)$, we determine λ_1 based on the steady state,

$$\lambda_1 = \frac{\mu_4[P^\infty]}{[D^\infty](K_1 - [D^\infty])} = (10^5, 10^7)\,(M^{-1}s^{-1}).$$

Parameters in Eq. 7. Trimers are formed from monomers and dimers, with the concentration around $10^{-6}\,(M)$. Since T arises from the combination of these species, its steady-state concentration, $[T^\infty]$, is greater than that of dimers, $[D^\infty]$. To reflect this, we set the carrying capacity K_2 within the range $(10^{-6}, 4 \times 10^{-6})\,(M)$, ensuring K_2 has a larger lower bound compared to K_1. This setting maintains the inequality $10^{-12} \le [T^\infty](K_2 - [T^\infty]) \le 10^{-11}$. Since $[D^\infty](K_1 - [D^\infty])$ is approximated up to $10^{-12}\,(M)$, we use $[T^\infty](K_2 - [T^\infty]) \approx 10^{-11}\,(M)$ to estimate a range for λ_2. Similar to P, we set $[N^\infty]$ between 10^{-4} and $10^{-3}\,(M)$ and by selecting $\mu_5 \approx 10^{-4}\,(s^{-1})$ at steady state, λ_2 is estimated as follows:

$$\lambda_2 = \frac{\mu_5[N^\infty]}{[T^\infty](K_2 - [T^\infty])} = (10^3, 10^4)\,(M^{-1}s^{-1}).$$

Parameters in Eq. 8. In this equation, K_3 represents the carrying capacity for N, and thus we take $K_3 = [N^\infty] \approx 10^{-3}$ (M). In the steady-state equation, we have

$$\frac{1}{2}\lambda_3[P^\infty] - \mu_6[F^\infty] = 0.$$

By setting $[P^\infty] = 10^{-4}$ (M), and assuming $\mu_6 = (5 \times 10^{-5}, 10^{-4})\,(s^{-1})$, we have λ_3 be within the range $[0, 1]\,(s^{-1})$. It is important to note that in this equation, we consider a normalized F with $[F^\infty] = 1\,(M)$ which has been widely considered in the literature[22]. In Table 1, we summarize the values and ranges for the parameters of equations (2)–(8).

Sensitivity Analysis

In this section, we employ Sobol sensitivity analysis, a variance-based methodology derived from the analysis of variance[29]. The Saltelli sampling approach, a derivative of Quasi-Monte Carlo methods, was employed for effective Sobol sensitivity analysis[30]. For simplicity, we denote our model as

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{\theta}),$$

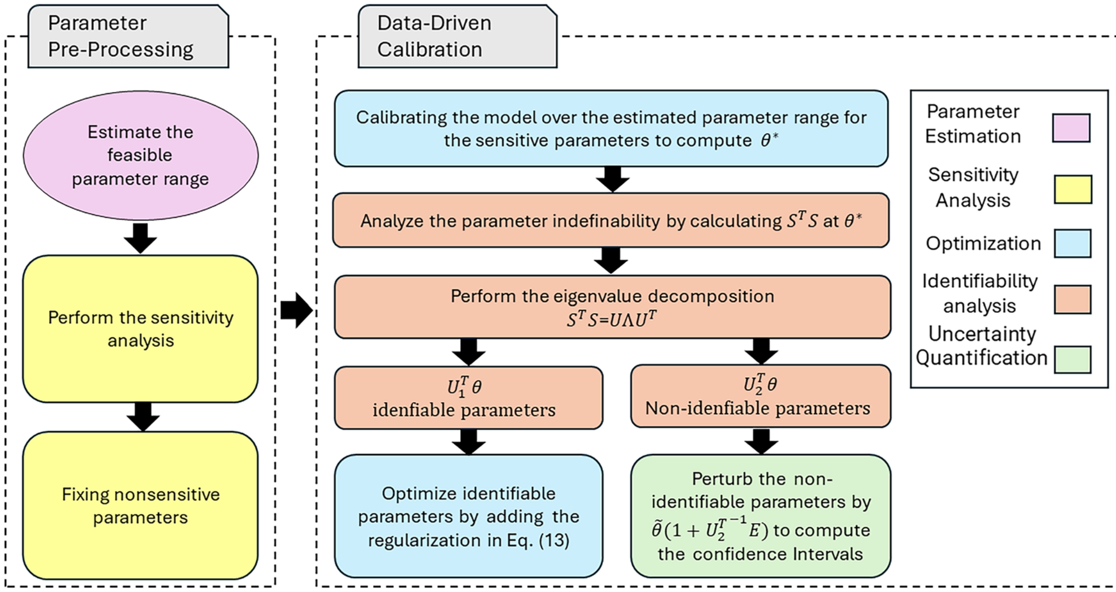


Fig. 12 | The flowchart of the data-driven modeling approach. Parameter pre-processing produces the feasible parameter range summarized in Table 1, and the sensitivity analysis identifies the non-sensitive parameters, as shown in Table 5. Our data-driven approach calibrates the sensitive parameters using experimental data. We also perform the parameter identifiability analysis to categorize these sensitive parameters into identifiable parameters, which can be calibrated using the experimental data, and non-identifiable parameters, which contribute to the model's uncertainty but cannot be identified using the experimental data.

where $\boldsymbol{x} = [M, D, T, P, N, F]$ represents the state variables, $\boldsymbol{\theta}$ denotes all the parameters, and $\boldsymbol{f}$ is the right-hand side of the system of differential equations.

Initially, we generated N samples for each parameter, forming the basis of our parameter space exploration. For each parameter, θ_i, we created two matrices: one varying θ_i while keeping other parameters fixed, and another varying all parameters except θ_i. This approach helps assess the individual and combined effects of parameters, essential for first and second-order Sobol sensitivity analyses. Using $N(2n_p + 2)$ sample points, where N is the base sample size and n_p is the number of parameters, the Saltelli method improves the convergence rate, leading to more precise results with fewer samples compared to traditional sampling.

Data-driven modeling

In our study, we utilize experimental data collected according to a well-established protocol for aggregating beta-amyloid (β-amyloid) peptides. These measurements were performed using Thioflavin T, a dye that fluoresces more as aggregate levels increase. The data collection and subsequent analyses adhere strictly to the methods outlined in[9]. Quality control measures are applied to ensure the reliability and reproducibility of the data. As highlighted in[22], this includes controls over experimental conditions such as protein purity and the validation of assumptions like the proportionality of dye fluorescence to aggregate concentrations. Following these rigorous standards is crucial not only for data accuracy but also for the validity of any conclusions drawn from the kinetic analysis. This meticulous approach allows us to extract meaningful data of fibrils into the mechanisms of peptide aggregation, underpinning the scientific value of our modeling efforts. This data includes the longitudinal concentration of fibrils, denoted as F in our model, for 10 different initial monomer concentrations. According to literature[9–11,22], it is a common approach to use normalized data for calibration. Therefore, we first normalize the available data set and map it between 0 and 1 such that the process starts from 0 as the initial value of F and reaches a steady state at $F(T) = 1$. Figure 11 shows the raw and normalized concentrations of $F(t)$ for different initial conditions of $M(0)$.

We outlined our data-driven approach in Figure 12.

The feasible parameter range is summarized in Table 1, and the sensitivity analysis identifies the non-sensitive parameters based on Figure 5.

Table 5 | Fixed values of non-sensitive parameters

Parameter	k_{21}	μ_2	μ_3	$\hat{\mu}_1$	ϵ_1	ϵ_2
Fixed Value	8×10^{-10}	9×10^{-9}	9×10^{-9}	1×10^{-14}	1×10^{-13}	1×10^{-11}

These parameters were determined based on prior studies or assumed to be negligible within the model's sensitivity range.

Thus we fix these non-sensitive parameters and choose the lower bound of these parameters as their fixed values, as given in Table 5.

Following these preprocessing procedures, in this section, we will calibrate our model using experimental data and perform parameter identifiability and uncertainty quantification.

Feasible parameter ranges for different initial monomer concentrations. Since some parameters, such as the carrying capacities for dimers and trimers, K_1 and K_2, highly depend on the initial monomer values, $M(0)$, we divide the initial values into two groups: $M(0) > 2\mu$M and $M(0) \leq 2\mu$M. The parameter ranges listed in Table 1 are suitable for initial values in the first group. However, for $M(0) \leq 2\mu$M, it is necessary to reduce the upper bound of the K_1 and K_2 ranges from 4×10^{-6} (M) to 2×10^{-6} (M). As a result, the range of λ_2 should also be updated.

We recall that in Section Parameters in Eq. (7), we used the inequality $10^{-12} \leq [T^\infty](K_2 - [T^\infty]) \leq 10^{-11}$ (M) and approximated this term with 10^{-11} (M) to find a range for λ_2 as $(10^3, 10^4)$ $(M^{-1} s^{-1})$. Changing the upper bound of K_2 from 4×10^{-6} (M) to 2×10^{-6} (M) requires using 10^{-12} (M) as the closer approximation value for $[T^\infty](K_2 - [T^\infty])$, which makes λ_2 fall within the range $(10^4, 10^5)$ $(M^{-1}s^{-1})$. Thus we summarize the three parameter values in Table 3.

Model calibration based on experimental data. The ranges specified in Tables 1 and 3 define a feasible region within the parameter space for $M(0) = 5, \ldots, 1.1$, $(\mu$M). To find an optimal parameter vector, θ^*, By using the normalized experimental data and Nelder-Mead optimization method[31], we solve the following optimization problem to obtain the optimal values for sensitive parameters:

$$\theta^* = \arg\min_{\boldsymbol{\theta}} \sum_{j=1}^{N} \left(F(t_j; \boldsymbol{\theta}) - F_j\right)^2, \tag{9}$$

where $F(t_j; \boldsymbol{\theta})$ represents the solution of Eq. (8) at time t_j and F_j is the normalized experimental data at t_j. This approach enables us to determine the parameter values, achieving an optimal match between the observed data and the simulated solution to F.

Parameter practical identifiability analysis. Next, we perform the practical identifiability analysis for all the parameters. As discussed in Supplementary Note 3, the parameter identifiability is related to the sensitivity matrix defined as

$$S = \begin{bmatrix} s_1(t_1) & s_2(t_1) & \dots & s_q(t_1) \\ s_1(t_2) & s_2(t_2) & \dots & s_q(t_2) \\ \vdots & \vdots & \ddots & \vdots \\ s_1(t_N) & s_2(t_N) & \dots & s_q(t_N) \end{bmatrix} \tag{10}$$

where $s_k(t_j) = \frac{\partial y(t_j, \theta^*)}{\partial \theta_k}$ and t_j is corresponding to the given data y_j. According to Supplementary Note 3, we decompose the S^TS matrix as

$$S^T S = U \Lambda U^T, \tag{11}$$

where U is an $n_p \times n_p$ matrix of eigenvectors, and Λ is a diagonal matrix corresponding to the eigenvalues, e.g. $\lambda_1 > \lambda_2 > \cdots > \lambda_r > 0$, where r is the rank. Thus we can split the non-zero and zero eigenvalues of S^TS as

$$S^T S = \begin{bmatrix} U_1 & U_2 \end{bmatrix} \begin{bmatrix} \Lambda_1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} U_1^T \\ U_2^T \end{bmatrix}, \tag{12}$$

where Λ_1 contains non-zero eigenvalues $\lambda_1, \cdots, \lambda_r$. In this case, we can decompose the parameters into two groups:

- **Identifiable Parameters**: These are defined as $U_1^T\theta$, corresponding to the non-zero eigenvalues. Specifically, as shown in Supplementary Note 3, we can express

$$\nabla L_\epsilon(\theta) = \Lambda_1 U_1^T(\theta - \theta^*) + O(\epsilon) + O(\| \theta - \theta^* \|^2) = 0,$$

  which means that we can solve $U_1^T\theta$ based on the optimization problem, making them identifiable.
- **Non-identifiable Parameters**: These are defined as $U_2^T\theta$, corresponding to the kernel of S^TS. These parameters cannot be identified by solving the optimization problem directly. To address this, we regularize the optimization problem by introducing a regularization term.

In practice, we determine the non-zero eigenvalues by setting a tolerance, e.g., $tr = 10^{-5}$, to set the identifiable and non-identifiable parameters. Eigenvectors associated with large eigenvalues $\lambda_i > tr$ indicate directions in the parameter space with significant variance, meaning these directions are well-determined by the data and are considered *identifiable*. Conversely, eigenvectors corresponding to small or near-zero eigenvalues, e.g., $\lambda_i < tr$, suggest that changes in these directions hardly affect the output. This implies that the data does not provide sufficient information to uniquely determine parameters in these directions. Therefore, any linear combination of parameters corresponding to these eigenvectors remains uncertain or *non-identifiable*.

Therefore, we attempt to enhance our calibration by considering the following minimization problem with an added regularization term:

$$\tilde{\theta} = \arg\min \left(\sum_{j=1}^{N} (y(t_j, \theta) - y_j)^2 + \lambda \, \| \, U_2^T\theta - U_2^T\theta^* \|^2 \right), \tag{13}$$

Here, the cost function consists of two parts. The first term measures the sum of squared differences between model predictions $y(t_j, \theta)$ at time t_j and observed data y_j. This term ensures that the model fits the data well and the regularization term is specifically designed to handle the issue of non-identifiability. The regularization term penalizes the difference between the projected values of the new parameter vector θ and the reference vector θ^* onto the subspace defined by U_2. The regularization parameter λ controls the balance between fitting the model to the data and penalizing large deviations from θ^*.After optimizing the parameter values for 10 different initial concentrations of $M(0)$, we summarized the statistics of these values in Fig. 6. The detailed values of $\tilde{\theta}$ are summarized in Supplementary Table 1 of Supplementary Note 4.

Uncertainty quantification. In this part, we assess the model uncertainty introduced by both the non-identifiable parameters and the initial condition. Specifically, we first perturb the initial condition of F by $F(0) = \epsilon$, where ϵ follows a normal distribution, $\mathcal{N}(0, 0.05)$. This perturbation quantifies the model uncertainty introduced by the initial value of $F(0)$. Secondly, to address uncertainties introduced by the non-identifiable parameters, we define a perturbation vector Ξ whose elements follow a normal distribution. The model parameters are then perturbed by:

$$\bar{\theta} = \tilde{\theta} \times (1 + (U_2^T)^{-1}\Xi), \tag{14}$$

where $(U_2^T)^{-1}$ is the pseudoinverse of U_2^T, projecting the perturbations onto the subspace defined by the non-identifiable directions.

By randomly generating 1000 perturbation vectors, we solved the model using the perturbed parameters and initial conditions and computed the 95% confidence interval for these solutions. In Fig. 7, the concentrations of all state variables for the initial condition $M(0) = 2\ \mu M$ are plotted using the optimal parameter vector $\tilde{\theta}$, along with their respective confidence intervals. Additionally, Figure 8 presents the confidence intervals, model solutions F, and observations for other initial conditions $M(0)$. This comprehensive analysis allows us to assess the impact of parameter uncertainties on the simulated solution's reliability under varied conditions influenced by the non-identifiable parameters.

Optimal treatment studies of anti-Aβ monoclonal antibodies

In recent years, significant advancements have been made in the therapeutic landscape of AD, particularly with the development of monoclonal antibody drugs such as donanemab, lecanemab, and aducanumab. These drugs aim to address the underlying amyloid pathology associated with AD by targeting Aβ aggregates in the brain, as schematically illustrated in Fig. 1.

- Donanemab: this drug specifically targets fibrillar plaques and fibrils. Donanemab binds to particular epitopes on the fibrils, facilitating their clearance through immune-mediated mechanisms, promoting the removal of these aggregates[32,33].
- Lecanemab: lecanemab primarily targets Aβ protofibrils, which are intermediate aggregates in the pathway to forming insoluble fibrils. This action helps prevent the progression to larger, more stable amyloid structures[32,34,35].
- Aducanumab: aducanumab has a broader target range, including soluble oligomers and insoluble fibrils. By binding to these aggregates, aducanumab activates microglial cells, enhancing the phagocytosis and clearance of amyloid plaques[36,37].These therapeutic strategies focus on targeting various stages of the Aβ aggregation cascade to mitigate AD progression and improve patient outcomes. In our model, these anti-Aβ therapies are introduced as control functions $u_D(t)$ (Donanemab),

$u_L(t)$ (Lecanemab), and $u_A(t)$ (Aducanumab), each representing the respective drug's effect on reducing amyloid burden. By incorporating these control functions into the state equations, we derive the following equations for P, N, and F, while the equations for monomers, dimers, and trimers Eqs. ((2)–(4)) remain unchanged. Together, these equations form the state system for the optimal control problem.

$$\begin{aligned}\frac{d[P]}{dt} &= \lambda_1[D](K_1-[D]) - \mu_4[P] - u_L(t)[P] - u_A(t)[P]\\ \frac{d[N]}{dt} &= \lambda_2[T](K_2-[T]) - \mu_5[N] - u_A(t)[N]\\ \frac{d[F]}{dt} &= \lambda_3\left(\frac{[P]}{1+[N]/K_3}\right) - \mu_6[F] - u_D(t)[F] - u_L(t)[F] - u_A(t)[F]\end{aligned} \tag{15}$$

The optimal anti-Aβ treatment seeks to reduce fibrils $F(t)$ and toxic aggregated species $N(t)$ while minimizing side effects. The objective function, which accounts for the effects of three treatments is defined as follows:

$$\begin{aligned}J(u_D,u_L,u_A) &= \gamma F(T) + \int_0^T\big(j_1N(t)+j_2F(t)\\ &\quad +\tfrac{1}{2}g(t)\left[\left(\tfrac{u_D(t)}{U^D_{\max}}\right)^2 + \left(\tfrac{u_L(t)}{U^L_{\max}}\right)^2 + \left(\tfrac{u_A(t)}{U^A_{\max}}\right)^2\right]\\ &\quad + c_DS_D(u_D)F(t) + c_LS_L(u_L)F(t) + c_AS_A(u_A)F(t)\big)\,dt.\end{aligned} \tag{16}$$

We seek to find an optimal control vector (u_D^*, u_L^*, u_A^*) to minimize our objective function,

$$J(u_D^*, u_L^*, u_A^*) = \min_{(u_D,u_L,u_A)} J(u_D,u_L,u_A).$$

The cost function addresses several objectives related to effective and safe treatment strategies. The first term, $\gamma F(T)$, minimizes the concentration of fibrils $F(T)$ at the end of the treatment period T, targeting the reduction of amyloid plaques. The integral terms $j_1N(t)$ and $j_2F(t)$ measure the total accumulation of toxic oligomers $N(t)$ and fibrils $F(t)$ over time. The quadratic terms $\left(\frac{u_D(t)}{U^D_{\max}}\right)^2$, $\left(\frac{u_L(t)}{U^L_{\max}}\right)^2$, and $\left(\frac{u_A(t)}{U^A_{\max}}\right)^2$, scaled by the time-dependent function $g(t)$, act as regularization terms that limit fluctuations and large magnitudes in the control variables $u_D(t)$, $u_L(t)$, and $u_A(t)$. The decay function $g(t)$ modifies the contribution of these terms over time, allowing the model to incorporate changes in treatment intensity or dosage.

The final terms $c_DS_D(u_D)F(t)$, $c_LS_L(u_L)F(t)$, and $c_AS_A(u_A)F(t)$ capture the potential side effects of each drug. Side effects are influenced by two factors: high drug dosage and the presence of larger amyloid plaques. This interaction is represented by the product of $S(u)$, the side effect function of the drug, and $F(t)$, the fibril concentration.

Our model focuses on ARIA-E, a subtype of ARIA, as the key measure of drug side effects because it is both common and closely linked to dosage. ARIA-E, which causes brain swelling or fluid buildup, tends to increase with higher drug doses, making it a useful marker for evaluating treatment risks. The controls are $L^\infty(0, T)$ functions with constraints:

$$0 \le u_D(t) \le U^D_{\max},\ 0 \le u_L(t) \le U^L_{\max},\ 0 \le u_A(t) \le U^A_{\max},\ \forall t \in [0, T].$$

In this section, we use clinical data to develop a dose-dependent side effect function. To ensure consistency, we focus on studies that report amyloid plaque reduction at the end of treatment, allowing us to estimate the maximum usable dosage in our mathematical model based on each drug's clearance rate. We specifically selected studies that not only align with FDA-approved package inserts but also include additional data on alternative treatment regimens, such as different dosage levels and their associated side effects. By incorporating a broader dataset, we aim to refine the side effect function and establish a more accurate estimation of the maximum dosage that can be realistically applied in our model. To solve this optimal control problem, we will analyze the effects of each drug individually. By isolating the impact of each treatment, Donanemab, Lecanemab, and Aducanumab, we aim to understand their specific contributions to therapeutic results.

Analysis of donanemab treatment. Now, we consider u_D first and set $u_L = u_A = 0$ in Eq. (15). In addition, we suppose that $g(t) = \alpha e^{-\beta t}$ and the cost function to be minimized becomes,

$$J(u_D) = \gamma F(T) + \int_0^T\left(j_1N(t) + j_2F(t) + \frac{1}{2}\alpha e^{-\beta t}\left(\frac{u_D(t)}{U^D_{\max}}\right)^2 + c_DS_D(u_D)F(t)\right)dt.$$

To derive the adjoint equations for this optimal control problem, we first introduce the Hamiltonian H[38]:

$$H(x(t), u(t), \Lambda(t)) = L(x(t), u(t)) + \Lambda(t)^\top f(x(t), u(t)),$$

where $L(x(t), u(t))$ represents the Lagrangian of the system, the cost function integrand, and $\Lambda(t)$ is the vector of adjoint variables. The term $f(x(t), u(t))$ represents the right-hand side of the system of ODEs, which includes the state variables M, D, T, P, N, F, along with the control functions. It is important to clarify that, although the ODE system was initially expressed as

$$\frac{dx}{dt} = f(x(t), \theta),$$

where θ represents the parameters, once these parameters are calculated and fixed, we include the control $u(t)$ in the optimal control problem. Therefore, the right-hand side is simplified to $f(x(t), u(t))$. To focus specifically on the effects of Donanemab, we set $u_L = 0$ and $u_A = 0$.

Using Pontryagin's Maximum Principle, the adjoint equations for this scenario are derived and detailed in Eq. (S13) in Supplementary Note 5. These equations follow the general form:

$$\frac{d\Lambda(t)}{dt} = -\frac{\partial H}{\partial \boldsymbol{x}},$$

where $\Lambda(t) = [\lambda_M, \lambda_D, \lambda_T, \lambda_P, \lambda_N, \lambda_F]^\top$ is the vector of adjoint variables. The boundary conditions for the adjoint variables are specified as:

$$\Lambda(T) = [0, 0, 0, 0, 0, \gamma]^\top,$$

where γ is a constant, and the adjoint variables corresponding to the state variables M, D, T, P, and N are zero at the final time T, with only $\lambda_F = \gamma$, being nonzero.

To estimate the maximum dosage of Donanemab ($U^D_{\max}$), we used data from the TRAILBLAZER-ALZ 2 trial[39], which included 1182 patients with early symptomatic AD in the low/medium tau subgroup. In this group, brain amyloid plaque levels-measured via PET imaging in Centiloid units-decreased from 102.4 to 14.4 over 76 weeks (a reduction of 88.0 Centiloids). The following equation describes the corresponding dynamics of amyloid plaques:

$$\frac{dA\beta}{dt} = \Lambda - \mu A\beta - U^D_{\max}A\beta, \tag{17}$$

where:

- Λ: Production of amyloid plaques from smaller species aggregating into plaques,
- $-\mu A\beta$: Natural clearance rate of Aβ,
- $-U^D_{\max}A\beta$: Drug-induced clearance of Aβ.

At equilibrium:

$$A\beta_I = \frac{\Lambda}{\mu},\ A\beta_F = \frac{\Lambda}{\mu + U^D_{\max}},$$

Table 6 | Baseline and final amyloid plaque sizes for different Lecanemab treatment regimens

Regimen	2.5 mg/kg Biweekly	5 mg/kg Monthly	5 mg/kg Biweekly	10 mg/kg Monthly	10 mg/kg Biweekly
Baseline	1.41	1.42	1.40	1.42	1.37
Final size	1.316	1.289	1.203	1.195	1.064

where $A\beta_I$ and $A\beta_F$ are the initial and final plaque levels, respectively. From clinical data:

$$\frac{A\beta_F}{A\beta_I} = \frac{14.4}{102.4} \approx 0.14.$$

Substituting this into the equilibrium equations:

$$\frac{\mu}{\mu + U^D_{\max}} = 0.14 \ \Rightarrow \ U^D_{\max} = \frac{86}{14}\mu.$$

We incorporate this relationship between $U^D_{\max}$ and the natural clearance rate into our model and consider $U^D_{\max} = \frac{43}{7}\mu_6$. We note that this model does not assume that the antibody is administered at the same dose, frequency, or schedule as in clinical trials. Instead, the clinical trial data is used solely to estimate $U_{\max}$, the maximum achievable drug-induced clearance rate. This defines an upper bound on therapeutic efficacy. The actual treatment schedule is governed by the control function $u(t)$, which is co-optimized by the model to find the most effective time-varying application of the drug.

Clinical data from[39] report Amyloid-Related Imaging Abnormalities-Edema (ARIA-E) in 24.0% of the Donanemab group and 1.9% of the placebo group, based on routine MRI monitoring throughout the trial. The probability of ARIA-E is modeled as proportional to the dosage $u_D(t)$:

$$S_D(u_D) = 0.221\,\frac{u_D(t)}{U^D_{\max}}.$$

This linear relationship links the side effects to the dosage administered during treatment.

Optimal control problem with estimated parameters. In this section, we set $j_1 = j_2 = c_D = 1$, $\alpha = 2$, and $\beta = 2 \times 10^{-4}$. The maximum dosage $U^D_{\max}$ and the side effect function $S_D(u_D)$ are used as previously estimated. The optimal control problem is then solved using the forward-backward sweep algorithm with boundary conditions. The algorithm iteratively adjusts the control $u_D(t)$ to reduce the cost function and achieve convergence to an optimal solution.

To solve the optimal control problem, we begin by initializing the control vector $u_D(t)$ to zero. The system of state differential equations is then solved forward in time using the initial conditions and the Runge-Kutta method to compute the state variables. These state variables are subsequently used to solve the adjoint equations backward in time, starting from the boundary conditions, to calculate the adjoint variables (Algorithm 1).

The optimal control $u^*_D(t)$ is derived from the Hamiltonian condition on the interior of the control set:

$$\frac{\partial H}{\partial u_D} = 0 \Rightarrow \frac{0.221 c_D F}{U^D_{\max}} + \alpha e^{-\beta t}\,\frac{u_D(t)}{(U^D_{\max})^2} - \lambda_F F = 0.$$

Taking the bounds into account, $0 \le u_D(t) \le U^D_{\max}$, leads to the following updated control:

$$u^D_{\text{new}}(t) = \min\left(U^D_{\max}, \max\left(0, (U^D_{\max})^2\,\frac{\lambda_F F - \frac{0.221 c_D F}{U^D_{\max}}}{\alpha e^{-\beta t}}\right)\right).$$

Table 7 | Relationship between $U^L_{\max}$ and μ_6 for each treatment group along with reported ARIA-E rates

Treatment Group	$U^L_{\max}$	ARIA-E (%)
Lecanemab 2.5 mg/kg biweekly	$\frac{7}{93}\mu_6$	1.9
Lecanemab 5 mg/kg monthly	$\frac{9}{91}\mu_6$	2.0
Lecanemab 5 mg/kg biweekly	$\frac{14}{86}\mu_6$	3.3
Lecanemab 10 mg/kg monthly	$\frac{16}{84}\mu_6$	9.9
Lecanemab 10 mg/kg biweekly	$\frac{22}{78}\mu_6$	9.9

The control is iteratively updated for stability and convergence using the following formula and this iterative process continues until $u_D(t)$ converges to a stable solution.

$$u_D(t) = 0.5\,u_D(t) + 0.5\,u^D_{\text{new}}(t).$$

We solve the optimal control system for each scenario by using ten different initial conditions for the monomer and their corresponding parameters, as determined in the previous sections. In Table 4, the row corresponding to treatment D presents the percentage reduction in fibril concentration, calculated as

$$\frac{F(T) - F_u(T)}{F(T)} \times 100.$$

Here, $F(T)$ represents the fibril concentration at the final time without any treatment ($u_D = 0$), while $F_u(T)$ denotes the fibril concentration at the same time in the presence of treatment ($u_D \neq 0$).

Analysis of Lecanemab Treatment. In this section, we focus exclusively on the effects of Lecanemab by setting $u_D = 0$ and $u_A = 0$ in Eq. (15), similar to our approach in the previous subsection, we minimize the following objective function,

$$J(u_L) = \gamma F(T) + \int_0^T \left(j_1 N(t) + j_2 F(t) + \frac{1}{2}\alpha e^{-\beta t}\left(\frac{u_L(t)}{U^L_{\max}}\right)^2 + c_L S_L(u_L) F(t)\right) dt.$$

and derive the adjoint equations for this case in Eq. (S14) given in Supplementary Note 5. For Lecanemab, $U^L_{\max}$, the maximum clearance rate of $A\beta$ plaques induced by the drug, can be calculated from clinical data given in[35].

In this study, a total of 856 patients with early AD were randomized across five Lecanemab treatment regimens: 2.5 mg/kg biweekly, 5 mg/kg monthly, 5 mg/kg biweekly, 10 mg/kg monthly, and 10 mg/kg biweekly.

Amyloid plaque burden was measured using PET imaging and quantified using standardized uptake value ratio (SUVR). The baseline SUVR values for these groups were reported as 1.41, 1.42, 1.40, 1.42, and 1.37, respectively. The corresponding mean changes from baseline after the treatment period were −0.094, −0.131, −0.197, −0.225, and −0.306, indicating dose-dependent plaque reduction.

Table 6 summarizes the baseline and final amyloid plaque sizes for these Lecanemab treatment regimens. The baseline values represent the initial amyloid plaque size before treatment, while the final values reflect levels after the full treatment course.

Using Eq. (17) and following a similar approach as for Donanemab, we derive a relationship between $U^L_{\max}$ and μ_6 for each treatment group. The results are summarized in Table 7, including the reported data for ARIA-E adverse effects for each treatment group as provided in the referenced paper.

To ensure a unique maximum dosage in our model and in the characteristic equation for u_L, we calculate the average of the maximum dosages

Table 8 | Baseline and Final Values for Aducanumab Treatment in the Emerge and Engage Groups

Group	Low Dose Engage	High Dose Engage	Low Dose Emerge	High Dose Emerge
Baseline	1.385	1.407	1.394	1.383
Final size	1.215	1.175	1.224	1.105
ARIA-E (%)	26	36	26	35

Table 9 | Estimated U^A_{max} and ARIA-E for Aducanumab

Treatment Group	U^A_{max} (μ_6)	ARIA-E (%)
Low Dose	$\frac{12}{88}\mu_6$	26
High Dose	$0.227\mu_6$	35.5

reported in this table and set $U^L_{\max} = 0.16\,\mu_6$. To maintain consistency in the side effect model across all treatments, similar to $S_D(t)$, we use the ARIA-E data provided in Table 7 to derive a linear fit for S_L as:

$$S_L(u_L) = 0.0563\,\frac{u_L(t)}{U^L_{\max}}.$$

Additionally, we consider $c_L = 1$ and substitute these terms into the optimal control model. Now, setting $\frac{\partial H}{\partial u_L} = 0$, on the interior of the control set and using the bounds, the characterization of the optimal control becomes:

$$u^L_{\text{new}}(t) = \min\left(U^L_{\max}, \max\left(0, (U^L_{\max})^2\,\frac{\lambda_P[P] + \lambda_F[F] - c_L\,0.0563\,\frac{F(t)}{U^L_{\max}}}{\alpha e^{-\beta t}}\right)\right).$$

The resulting model is then solved numerically using the forward-backward sweep algorithm with boundary conditions. In Table 4, the row corresponding to treatment L reflects the percentage reduction in fibril concentration, which is achieved exclusively through the control u_L, without any influence from u_D or u_A.

Analysis of Aducanumab treatment. In this section, we analyze the effects of Aducanumab by setting $u_D = 0$ and $u_L = 0$ in Eq. (15). Following the approach outlined in the previous subsections, we define the objective function for this treatment as:

$$J(u_A) = \gamma F(T) + \int_0^T \left(j_1 N(t) + j_2 F(t) + \frac{1}{2}\alpha e^{-\beta t}\left(\frac{u_A(t)}{U^A_{\max}}\right)^2 + c_A S_A(u_A) F(t)\right) dt.$$

We then derive the adjoint equations for this case, as detailed in Eq. (S15) in Supplementary Note 5.

Aducanumab's maximum dosage ($U^A_{\max}$) reflects the highest clearance rate of Aβ plaques facilitated by the drug. To estimate $U^A_{\max}$ for Aducanumab, we refer to the results reported in a clinical study[40]. 3285 patients with early AD were grouped into two categories: Emerge and Engage. The study included two dosage levels:

- Low Dose Group: Participants received either 3 or 6 mg/kg every 4 weeks over 76 weeks, with an ARIA-E incidence of 26%.

- High Dose Group: Participants received 10 mg/kg every 4 weeks over 76 weeks, with an ARIA-E incidence of 35%.

We summarize the data for amyloid PET SUVR values and ARIA-E incidence in the Emerge and Engage groups in Table 8.

The table captures essential data for both dosage levels across the Emerge and Engage groups. Considering the mean values between these groups, we calculate $U^A_{\max}$ based on μ_6, following a similar approach as in previous cases. Table 9 reports the results. To ensure a unique maximum dosage in our model and the characteristic equation for u_A, we calculate the average of the maximum dosages reported in this table and set $U^A_{\max} = 0.18\,\mu_6$. For Aducanumab, we use the ARIA-E data provided in Table 9. The side-effect function is defined as:

$$S_A(u_A) = 0.301\,\frac{u_A(t)}{U^A_{\max}}.$$

By setting $\frac{\partial H}{\partial u_A} = 0$ on the interior of the control set, we derive the optimal control $u_A(t)$ on that set. Taking the bounds into account, $0 \le u_A(t) \le U^A_{\max}$, the characterization of the optimal control becomes:

$$u^A_{\text{new}}(t) = \min\left(U^A_{\max}, \max\left(0, (U^A_{\max})^2\,\frac{\lambda_P[P] + \lambda_N[N] + \lambda_F[F] - c_A\,0.301\,\frac{F(t)}{U^A_{\max}}}{\alpha e^{-\beta t}}\right)\right).$$

The positivity of the control function is maintained by selecting $c_A = 0.1$ in this case. The optimal control problem for Aducanumab is then solved, and the percentage reduction in fibril concentration achieved only by this drug ($u_A \neq 0$ with $u_D = u_L = 0$) is reported in the entry labeled as treatment A in Table 4.

Analysis of ALL treatments. To analyze the effectiveness of different treatments, we derive the adjoint equations Eq. (S16) by using Eq. 16 as the cost function and Eq. 15 as the state equations. Then, We used these equations to solve the optimal control problem for various treatment scenarios by specifying which controls (u_D, u_L, u_A) are active. In all cases, we use the same $U_{\max}$ value as in single-drug treatments because no independent clinical data exist to define a separate maximum effect for combinations. Even if such data were available, attributing observed plaque reduction to individual drugs would be underdetermined without pharmacodynamic details. This assumption preserves model identifiability and interpretability while remaining consistent with available data.The results are summarized in Table 4, which shows the percentage reduction in fibril concentration (F) for different initial values of $M(0)$ under various treatment combinations.

In Table 4, the "All Drugs" scenario considers all three treatments simultaneously. Here, all controls are active (u_D, u_L, $u_A \neq 0$), and the reductions in F reflect the outcome of solving the full system, including all terms in the adjoint equations, cost function, and state equations.

The "L and A" scenario evaluates fibril reduction when Donanemab is excluded ($u_D = 0$), leaving only Lecanemab and Aducanemab (u_L and u_A) active. Similarly, the "A and D" scenario examines the effect of combining Aducanemab and Donanemab (u_A and u_D), with Lecanemab set to zero ($u_L = 0$). Finally, the "L and D" scenario explores the combination of Lecanemab and Donanemab (u_L and u_D), excluding Aducanemab ($u_A = 0$).

Figure 9 shows the optimal control functions for Donanemab (u_D), Aducanumab (u_A), and Lecanemab (u_L) across all initial monomer concentrations $M(0)$, for both single and combination treatment scenarios. Solid and dashed lines are used to compare individual versus combined therapy, and the control strategies are obtained by solving the corresponding optimal control problems as described above.

We also compute the 95% confidence intervals for fibril concentrations $F(t)$ under ten different initial monomer concentrations $M(0)$, employing the same perturbation method for non-identifiable parameters outlined in Section Uncertainty Quantification, as depicted in Fig. 10.

Algorithm 1. Forward-Backward Sweep Algorithm with Boundary Conditions

1: Initialize the control $u_D(t)$ as a zero function.

2: while the relative error is not sufficiently small do

3: Solve the state equations forward in time using the current control $u_D(t)$ to compute the state variables.

4: Solve the adjoint equations backward in time using the current states and control $u_D(t)$ to compute the adjoint variables.

5: Update the control function $u_D(t)$ using the optimal control characterization:

$$u_D(t) = \min\left(U^D_{\max}, \max\left(0, (U^D_{\max})^2 \frac{\lambda_F[F] - \frac{0.221 c_D[F]}{U^D_{\max}}}{\alpha_D e^{\beta_D t}}\right)\right).$$

6: Compute the relative error of the control and check for convergence.
7: end while

Data Availability

All data supporting the findings of this study are available from the original study by[22] and its Supplementary Information can be accessed at https://www.nature.com/articles/nprot.2016.010. The code supporting the findings of this study are available on GitHub at https://github.com/LRabiei/AD-modeling-and-treatment.

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References

1. Prince, M. et al. World alzheimer report 2015: The global impact of dementia - an analysis of prevalence, incidence, cost and trends. *Alzheimer's Disease International (ADI)* https://www.alzint.org/resource/world-alzheimer-report-2015/. (2015).
2. Fraser, P. E., Duffy, L. K., Nguyen, J., Inouye, H. & Kirschner, D. A. Morphology and antibody recognition of synthetic β-amyloid peptides. *J. Neurosci. Res.* **28**, 474–485 (1991).
3. Selkoe, D. J. Alzheimer's disease: genes, proteins, and therapy. *Physiol. Rev.* **81**, 741–766 (2001).
4. Hardy, J. & Selkoe, D. J. The amyloid hypothesis of alzheimer's disease: progress and problems on the road to therapeutics. *Science* **297**, 353–356 (2002).
5. Querfurth, H. W. & LaFerla, F. M. Alzheimer's disease. *N. Engl. J. Med.* **362**, 329–344 (2010).
6. Haass, C. & Selkoe, D. J. Soluble protein oligomers in neurodegeneration: lessons from the alzheimer's amyloid beta-peptide. *Nat. Rev. Mol. Cell Biol.* **8**, 101–112 (2007).
7. Masters, C. L. & Selkoe, D. J. Biochemistry of amyloid β-protein and amyloid deposits in alzheimer disease. *Cold Spring Harb. Perspect. Med.* **2**, a006262 (2012).
8. Jeremic, D., Jiménez-Díaz, L. & Navarro-López, J. D. Past, present and future of therapeutic strategies against amyloid-β peptides in alzheimer's disease: a systematic review. *Ageing Res. Rev.* **72**, 101496 (2021).
9. Cohen, S. I. A. et al. Proliferation of amyloid-β_{42} aggregates occurs through a secondary nucleation mechanism. *Proc. Natl. Acad. Sci. USA* **110**, 9758–9763 (2013).
10. Cohen, S. I., Vendruscolo, M., Dobson, C. M. & Knowles, T. P. From macroscopic measurements to microscopic mechanisms of protein aggregation. *J. Mol. Biol.* **421**, 160–171 (2012).
11. Linse, S. Monomer-dependent secondary nucleation in amyloid formation. *Biophys. Rev.* **9**, 329–338 (2017).
12. Bertsch, M., Hilhorst, D., Marciniak-Czochra, A. & Zubelli, J. P. Multiscale modeling of amyloid-beta dynamics and inter-neuronal propagation. *J. Theor. Biol.* **400**, 123–134 (2016).
13. Fornari, E. et al. Spatial dynamics in amyloid-beta and tau propagation. *NeuroImage* **215**, 116824 (2020).
14. Hao, W. & Friedman, A. Mathematical model on alzheimer's disease. *BMC Syst. Biol.* **10**, 108 (2016).
15. Petrella, J. R., Hao, W., Rao, A., Doraiswamy, P. M. & Li, S. Computational causal modeling of the dynamic biomarker cascade in alzheimer's disease. *Comput. Math. Methods Med.* **2019**, 6216530 (2019).
16. Gellermann, G. P. et al. Abeta-globulomers are distinct pathological entities in alzheimer's disease. *J. Neurochem.* **107**, 1113–1124 (2008).
17. Garai, K. & Frieden, C. Quantitative analysis of the time course of amyloid-β oligomerization and subsequent growth of fibrils. *Biochemistry* **51**, 7406–7414 (2012).
18. Saccomani, M. P., Audoly, S. & D'Angio, G. Parameter identifiability in nonlinear systems: the role of initial conditions. *Automatica* **39**, 619–632 (2003).
19. Hao, W., Lenhart, S. & Petrella, J. R. Optimal anti-amyloid-beta therapy for alzheimer's disease via a personalized mathematical model. *PLoS Comput. Biol.* **18**, e1010481 (2022).
20. Ferreira, L. C. C., Rangan, A. V. & Durand, D. J. Design of optimal nonlinear network controllers for alzheimer's disease. *PLOS Comput. Biol.* **14**, e1006136 (2018).
21. Linse, S. et al. Kinetic fingerprints differentiate the mechanisms of action of anti-abeta antibodies. *Nat. Struct. Mol. Biol.* **27**, 1125–1133 (2020).
22. Meisl, G. et al. Molecular mechanisms of protein aggregation from global fitting of kinetic models. *Nat. Protoc.* **11**, 252–272 (2016).
23. Chiti, F. & Dobson, C. M. Protein misfolding, functional amyloid, and human disease. *Annu. Rev. Biochem.* **75**, 333–366 (2006).
24. Jeremic, D., az, L. & pez, J. D. peptides in Alzheimer's disease: a systematic review. *Ageing Res. Rev.* **72**, 101496 (2021).
25. Garai, K. & Frieden, C. Quantitative analysis of the time course of *aβ* oligomerization and subsequent growth steps using tetramethylrhodamine-labeled *aβ*. *Proc. Natl. Acad. Sci. USA* **110**, 3321–6 (2013).
26. Livadiotis, G., Ackleh, A. S., Elaydi, S. & Veprauskas, A. A continuous-time mathematical model and discrete approximations for the aggregation of β-amyloid. *J. Biol. Dyn.* **15**, 109–136 (2021).
27. Lindstrom, M. R., Chavez, M. B., Gross-Sable, E. A., Hayden, E. Y. & Teplow, D. B. From reaction kinetics to dementia: a simple dimer model of alzheimer's disease etiology. *PLOS Comput. Biol.* **17**, 1–24 (2021).
28. Craft, D., Wein, L. & Selkoe, D. A mathematical model of the impact of novel treatments on the *Aβ* burden in the Alzheimer's brain, CSF and plasma. *Bull. Math. Biol.* **64**, 1011–1031 (2002).
29. Sobol, I. Global sensitivity indices for nonlinear mathematical models and their monte carlo estimates. *Math. Comp. Simul.* **55**, 271–280 (2001).
30. Saltelli, A. Making best use of model evaluations to compute sensitivity indices. *Comp. Phys. Commun.* **145**, 280–297 (2002).
31. Nelder, J. A. & Mead, R. A simplex method for function minimization. *Comp. J.* **7**, 308–313 (1965).
32. Vogt, A. S., Jennings, G. T., Mohsen, M. O., Vogel, M. & Bachmann, M. F. Alzheimer's disease: a brief history of immunotherapies targeting amyloid beta. *Int. J. Mol. Sci.* **24**, 3895 (2023).
33. Söderberg, L., Johannesson, M. & Gkanatsiou, E. et al. Amyloid-beta antibody binding to cerebral amyloid angiopathy fibrils and risk for amyloid-related imaging abnormalities. *Sci. Rep.* **14**, 10868 (2024).
34. Johannesson, M. et al. Lecanemab demonstrates highly selective binding to abeta protofibrils isolated from alzheimer's disease brains. *Mol. Cell. Neurosci.* **130**, 103949 (2024).
35. Swanson, C. J. et al. A randomized, double-blind, phase 2b proof-of-concept clinical trial in early alzheimer's disease with lecanemab, an anti-*aβ* protofibril antibody. *Alzheimers Res. Ther.* **13**, 80 (2021).
36. Gunawardena, I. P. C., Retinasamy, T. & Shaikh, M. F. Is aducanumab for LMICs promises and challenges. *Brain Sci.* **11**, 1547 (2021).
37. Sevigny, J. et al. The antibody aducanumab reduces *aβ* plaques in Alzheimer's disease. *Nature* **537**, 50–56 (2016). Update in: Nature. **546**, 7659 (2017).
38. Lenhart, S. & Workman, J. T.*Optimal Control Applied to Biological Models*. Mathematical and Computational Biology Series (Chapman & Hall/CRC, Boca Raton, FL, 2007).

39. Sims, J. R. et al. Donanemab in early symptomatic alzheimer disease: The trailblazer-alz 2 randomized clinical trial. *JAMA* **330**, 512–527 (2023).
40. Budd Haeberlein, S. et al. Two randomized phase 3 studies of aducanumab in early alzheimer's disease. *J. Prev. Alzheimer's. Dis.* **9**, 197–210 (2022).

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Author contributions

K.R. developed and validated the models, performed sensitivity and identifiability analyses, parameter identification, and uncertainty quantification, and drafted the manuscript. J.R.P. contributed to the development of treatment strategies, offering guidance on optimal dosing regimens, addressing potential side effects, and collaborating onrefining therapeutic frameworks. S.L. advanced the application of optimal control theory in the study, providing critical insights that enhanced itsimplementation and aligned it with experimental considerations. C.L. contributed to improving the modeling ofaggregation, assisting in the development and refinement of specific model components. W.H. conceived the idea, conceptualized the study, supervised the work, guided the project's progress, provided detailed feedback, and editedthe manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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for the Alzheimer Disease Neuroimaging Initiative

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